GLoP-SAM3: A Dual Postprocessing Framework With Global–Local Prior Fusion and Reliability Gating for Open-Vocabulary Remote Sensing Segmentation

Chang Liu, Yuzhuo Han, Xiangzhe Kong, Mingjie Li, and Yonghua Li

***Abstract*—Open-vocabulary remote sensing semantic segmentation enables pixel-level recognition of arbitrary semantic categories through text prompts, providing an effective way to reduce the dependence of traditional closed-set remote sensing segmentation on fixed category sets and large-scale annotations. However, existing training-free methods in remote sensing scenarios still mainly rely on local responses and sliding-window inference and thus suffer from insufficient utilization of full-image global semantic priors, unstable local predictions, and unreliable prior propagation. To address these issues, we propose GLoP-SAM3, a dual postprocessing framework for open-vocabulary remote sensing semantic segmentation, whose core idea is to enhance performance through two lightweight postprocessing modules rather than introducing a new backbone or complex trainable components. Specifically, we develop a divergence-aware global–local fusion (DGLF) module, which adaptively integrates full-image semantic priors with local predictions by jointly modeling local–global discrepancy and local confidence, and a reliability-gated prediction refinement (RGPR) module, which suppresses erroneous prior propagation and improves segmentation quality in complex and boundary regions through high-confidence preservation, ambiguous-region enhancement, and prior reliability gating. Built upon SAM3 without any additional training, the proposed method introduces only limited computational overhead while significantly improving prediction consistency and segmentation accuracy. Extensive experiments on eight public remote sensing datasets demonstrate that GLoP-SAM3 consistently improves the baseline and achieves state-of-the-art (SOTA) performance under the training-free setting on representative benchmarks such as OpenEarthMap, validating the favorable balance of the proposed dual postprocessing strategy between efficiency and performance. The code, fixed prompts, and configuration files are available at https://github.com/LiorCohen0901/GLoP-SAM**

***Index Terms*—Open-vocabulary semantic segmentation (OVSS), postprocessing, remote sensing, SAM3, training-free.**

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I. Introduction

With the continuous improvement of remote sensing image acquisition capabilities, semantic segmentation has played an important role in land-cover mapping, urban scene understanding, environmental monitoring, and disaster assessment [1], [2], [3], [4]. Traditional remote sensing semantic segmentation methods are usually built upon closed-set category definitions and large-scale pixel-level annotations, which lead to obvious limitations in category scalability and cross-scene generalization [5], [6]. In contrast, open-vocabulary semantic segmentation (OVSS) enables the prediction of open-set categories through text prompts, providing a new perspective for flexible and scalable semantic understanding in complex remote sensing scenes. Furthermore, the training-free setting avoids additional training and large annotation costs, thus offering higher practical value in open environments [7], [8].

In recent years, a number of OVSS methods based on vision-language models have been proposed in the natural image domain [9], [10], [11], [12], and have achieved promising performance without additional training or with only limited training [13]. For remote sensing scenarios, methods such as SegEarth-OV and SegEarth-OV3 have also introduced open-vocabulary segmentation into high-resolution remote sensing image understanding, demonstrating the feasibility of this direction [14]. However, due to the characteristics of remote sensing images, such as large fields of view, complex object distributions, dense small targets, and high visual similarity among categories, existing methods are still easily affected by unstable local responses, semantic inconsistency across windows, and category ambiguity [12], [15]. In particular, for training-free methods such as SegEarth-OV3, the prediction results still mainly rely on local responses and sliding-window inference, lacking effective full-image global semantic priors, as well as explicit control over prior reliability and pixel-wise uncertainty. As illustrated in Fig. 1, existing training-free open-vocabulary remote sensing segmentation methods often suffer from incomplete local predictions and unreliable semantic responses in complex scenes, which limits their global semantic consistency and boundary quality. The proposed GLoP-SAM3 addresses these issues through a dual

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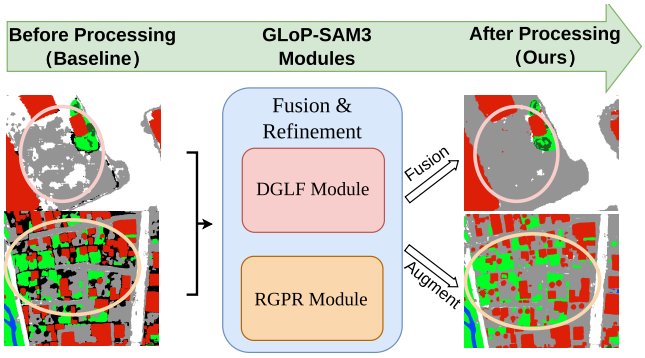


Fig. 1. Motivation illustration of the proposed GLoP-SAM3. Existing training-free open-vocabulary remote sensing segmentation methods often suffer from incomplete local predictions and semantic ambiguity, while the proposed DGLF and RGPR modules progressively improve prediction consistency and refinement quality.

postprocessing design, which progressively improves the prediction results by global–local fusion and reliability-guided refinement.

As a recently proposed foundation model, SAM3 provides stronger support for open-vocabulary perception and can simultaneously output instance-level and semantic-level responses, offering a new basis for building high-quality training-free open-vocabulary segmentation frameworks. However, when SAM3 is directly applied to open-vocabulary remote sensing semantic segmentation, its predictions still mainly depend on local responses, making it difficult to achieve effective collaboration between global and local information. Moreover, even when full-image priors are introduced, the absence of reliability control may lead to overcorrection in high-confidence regions and the propagation of erroneous priors in ambiguous areas. Therefore, how to effectively exploit global semantic priors while suppressing unreliable prior propagation, without introducing any additional training process, becomes a key issue for improving training-free open-vocabulary remote sensing segmentation.

It should be noted that the proposed framework is different from generic confidence-gated fusion or prior-refinement heuristics. Recent studies have explored related ideas from different perspectives, including local–global context modeling in LOGCAN++ [16], SAM-based segmentation enhancement [17], global–local feature refusion in GLFRNet [18], uncertainty-aggregated fusion in UAGLNet [19], and prior-guided semisupervised segmentation [20]. However, these methods are mainly designed for supervised or semisupervised closed-set segmentation, task-specific refinement, or mask-level enhancement, and they usually rely on fixed category spaces, task-specific training data, or learnable fusion/refinement networks. In contrast, GLoP-SAM3 is designed for the training-free open-vocabulary remote sensing setting, where both local predictions and full-image semantic priors are generated from SAM3 under text-defined categories. The proposed framework explicitly measures the distributional discrepancy between local predictions and global priors using Jensen–Shannon divergence, and further regulates prior injection through local confidence, ambiguous-region enhancement, and global prior reliability gating. Therefore, GLoP-SAM3 performs reliability-aware global–local prior reasoning rather than simple confidence-thresholding or generic prior refinement.

Motivated by the above observations, we propose GLoP-SAM3, a dual postprocessing framework for open-vocabulary remote sensing semantic segmentation. Built upon SAM3 as the basic predictor [21], GLoP-SAM3 performs reliability-aware global–local prior reasoning by integrating full-image semantic priors with local predictions and refining uncertain regions through reliability gating. It improves prediction consistency and segmentation accuracy without additional training, learnable fusion networks, or modifications to the SAM3 backbone. The main contributions of this article are summarized as follows.

1) We design dedicated preprocessing strategies for the global and local branches, respectively, so as to enhance the complementarity between full-image priors and local predictions and provide more stable inputs for subsequent fusion.
2) We propose a divergence-aware global–local fusion (DGLF) module, which adaptively integrates full-image semantic priors with local predictions by jointly modeling local–global discrepancy and local confidence, thereby improving the overall consistency and stability of prediction results.
3) We propose a reliability-gated prediction refinement (RGPR) module, which suppresses unreliable global prior propagation through high-confidence region preservation, ambiguous-region enhancement, and prior reliability gating, thus further improving the segmentation quality of complex and boundary regions [22], [23].
4) Based on the above two postprocessing modules, we construct a training-free framework for open-vocabulary remote sensing map segmentation, which effectively improves global semantic consistency and local prediction quality without significantly increasing computational overhead. Extensive experiments on eight public remote sensing datasets demonstrate the effectiveness of the proposed method, and competitive results are achieved on representative benchmarks such as OpenEarthMap.

The remainder of this article is organized as follows. Section II reviews related work. Section III presents the theoretical details of the proposed framework. Section IV reports the experimental results and analysis. Finally, Section V concludes this article and discusses future work.

II. Related Work

In recent years, OVSS has gradually become an important research topic in visual understanding, aiming to achieve pixel-level recognition of arbitrary text-defined categories without relying on a fixed category set. Existing studies on this task can be broadly divided into training-based methods, training-free methods, and OVSS methods for remote sensing scenarios. The first two lines of work are mainly developed in the

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natural image domain and provide the fundamental technical paradigms for open-vocabulary segmentation, while the third further extends these ideas to high-resolution remote sensing image understanding. In the following, we review related work from these three perspectives and analyze their main advantages and limitations.

A. Training-Based OVSS

Training-based OVSS methods usually adapt pretrained vision-language models to pixel-level tasks using annotated datasets. Existing methods can be broadly divided into region-adaptation-based, dense matching and cost-aggregation-based, and context-calibration-based approaches [24], [25], [26].

Among region-adaptation-based methods, OVSeg [27] improves region-level recognition by adapting CLIP to masked regions. MaskCLIP [28] further incorporates mask tokens into the ViT-based CLIP backbone to build a unified open-vocabulary universal image segmentation framework, while MasQCLIP [29] improves adaptation efficiency through MasQ-Tuning. Moving beyond region-level recognition, CAT-Seg [30] models finer-grained pixel-text matching relationships through cost aggregation in both spatial and class dimensions. SCAN [31] further enhances this line by strengthening the alignment between visual content and open-vocabulary semantic concepts, thereby alleviating the influence of insufficient global context and background noise. Overall, training-based methods have significantly improved open-vocabulary segmentation performance, but their dependence on additional training and adaptation limits their flexibility in open-world deployment.

B. Training-Free OVSS

Training-free methods avoid additional training by directly adapting pretrained vision-language models for dense prediction [11], [12]. Early studies mainly focused on improving the local discriminability of CLIP itself. NACLIP [32] enhances neighborhood locality in self-attention, ClearCLIP [33] reduces global bias and noisy components in CLIP representations, and SC-CLIP [34] further improves spatial discriminability through token-level anomaly correction and self-calibration.

More recent studies have shifted toward patch correlation modeling and semantic consistency. CorrCLIP [35], for example, identifies inconsistent interpatch correlations as a key bottleneck and reconstructs more coherent patch correlations to improve pixel-level prediction consistency. In general, training-free methods have evolved from simple CLIP locality enhancement to more explicit modeling of patch correlation and semantic consistency. However, most of them are primarily designed for natural images and still suffer from unstable local predictions, insufficient global semantic consistency, and unreliable prior propagation in more complex scenarios.

C. Open-Vocabulary Remote Sensing Semantic Segmentation

Extending OVSS to remote sensing imagery is particularly challenging due to extreme scale variation, arbitrary object orientation, and highly complex spatial layouts. Therefore, related studies have gradually moved from relying solely on CLIP semantics to leveraging stronger segmentation and detection foundation models for better spatial priors.

In this process, models such as DINO [36] and the SAM family [37], [38], [39] have played an important role by providing stronger localization ability and prompt-driven segmentation capability [40], [41], [42]. However, such combination-based pipelines often suffer from weak coupling between modules, semantic-geometric inconsistency, and error propagation in dense remote sensing scenes. Under this background, SegEarth-OV [9] and SegEarth-OV3 [10] further extend open-vocabulary segmentation to remote sensing scenarios. In particular, SegEarth-OV3 [10] introduces SAM3 as a unified concept segmentation backbone and demonstrates its potential for remote sensing open-vocabulary segmentation. Nevertheless, although it alleviates localization and mask quality issues to some extent, its prediction results still mainly rely on local responses and patch-level processing, while full-image global semantic priors, prior reliability, and pixel-wise uncertainty remain insufficiently modeled. This directly motivates the proposed dual postprocessing framework, GLoP-SAM3.

III. Methodology

In this section, we introduce the proposed method for open-vocabulary remote sensing semantic segmentation. As illustrated in Fig. 2, the proposed GLoP-SAM3 framework consists of a global prior branch, a local prediction branch, a DGLF module, and an RGPR module. The main objective of our framework is to improve local prediction stability and global semantic consistency by effectively integrating SAM3-based open-vocabulary prediction, full-image prior modeling, discrepancy-aware fusion, and reliability-gated refinement. Specifically, we first generate class-level local predictions from open-vocabulary text prompts, then construct a global prior from the full-image perspective, and further perform adaptive fusion according to the discrepancy between local prediction and global prior. Finally, a reliability-gated refinement strategy is applied to suppress unnecessary or unreliable prior injection. The details of the overall frame work and each key component are presented in the Sections III-A–III-D.

A. Overview

Given an input remote sensing image $I \in \mathbb{R}^{H\times W\times 3}$ and an open-vocabulary text set $\mathcal{T} = \{t_1, t_2, \ldots, t_Q\}$, where H and W denote the image height and width, respectively, and Q denotes the number of text queries, our goal is to predict a pixel-wise semantic label map

$$Y \in \{0, 1, \ldots, C-1\}^{H\times W} \tag{1}$$

where C is the number of semantic classes.

Since the same semantic category may correspond to multiple textual expressions in the open-vocabulary setting, we define a mapping function

$$\phi(q) \in \{0, 1, \ldots, C-1\} \tag{2}$$

and denote by

$$\mathcal{Q}_c = \{q \mid \phi(q) = c\} \tag{3}$$

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the set of all text queries corresponding to class c.

For each text query t_q, SAM3 produces a query-level response map

$$\mathbf{S}^{(q)} \in \mathbb{R}^{H\times W}. \tag{4}$$

Based on the above mapping, multiple synonymous queries are aggregated into a class-level prediction map

$$\mathbf{L}_c(x,y) = \max_{q\in\mathcal{Q}_c} \mathbf{S}^{(q)}(x,y). \tag{5}$$

Accordingly, the class-level prediction tensor is denoted by

$$\mathbf{L} \in \mathbb{R}^{C\times H\times W}. \tag{6}$$

To incorporate full-image semantic context, we further construct a global prior map

$$\mathbf{G} \in \mathbb{R}^{C\times H\times W}. \tag{7}$$

Based on $\mathbf{L}$ and $\mathbf{G}$, we perform divergence-aware dynamic fusion and reliability-gated refinement to obtain the final fused class response map

$$\hat{\mathbf{L}} = \mathcal{F}(\mathbf{L}, \mathbf{G}) \tag{8}$$

where $\mathcal{F}(\cdot)$ denotes the proposed global–local collaborative fusion function. The final segmentation result is given by

$$\hat{Y}(x,y) = \arg\max_c \hat{\mathbf{L}}_c(x,y). \tag{9}$$

Overall, built upon the open-vocabulary prediction capability of SAM3, the proposed framework improves semantic consistency and prediction stability in remote sensing scenes through global prior modeling and subsequent fusion refinement.

B. SAM3-Based Open-Vocabulary Prediction

To obtain pixel-wise initial predictions under the open-vocabulary setting, we first model the response for each text prompt using SAM3. For a given text query t_q, SAM3 produces both instance-level and semantic-level responses. Specifically, let $\left\{\mathbf{M}_k^{(q)}, s_k^{(q)}\right\}_{k=1}^{K_q}$ denote the outputs of the instance branch for the qth query, where $\mathbf{M}_k^{(q)} \in \mathbb{R}^{H\times W}$ is the mask logit of the kth instance, $s_k^{(q)}$ is its corresponding confidence score, and K_q is the number of detected instances. Meanwhile, the semantic branch produces a semantic response map $\mathbf{A}^{(q)} \in \mathbb{R}^{H\times W}$, and the presence branch outputs a presence score $r^{(q)}$.

By combining the outputs of the instance, semantic, and presence branches, the final response map for the qth text query is defined as

$$\mathbf{S}^{(q)} = r^{(q)} \cdot \max\left(\mathbf{A}^{(q)}, \max_{1\le k\le K_q}\left(s_k^{(q)}\mathbf{M}_k^{(q)}\right)\right) \tag{10}$$

where the inner maximization selects the strongest instance-level response at each pixel location, and the outer maximization performs pixel-wise fusion between the instance-level and semantic-level responses. Finally, the presence score $r^{(q)}$ is used to recalibrate the overall response, suppressing spurious activations that are inconsistent with the current text query.

Since the same semantic category may correspond to multiple textual expressions in the open-vocabulary setting, we further aggregate query-level predictions into class-level predictions. Let $\phi(q)$ denote the class index of the qth text query, and let $\mathcal{Q}_c$ denote the set of all text queries belonging to class c. The class-level response is then defined as

$$\mathbf{L}_c(x,y) = \max_{q\in\mathcal{Q}_c} \mathbf{S}^{(q)}(x,y) \tag{11}$$

where $\mathbf{L}_c(x,y)$ denotes the prediction response of pixel (x,y) for class c. Accordingly, the class-level prediction tensor is denoted by $\mathbf{L} \in \mathbb{R}^{C\times H\times W}$.

Through the above process, the open-vocabulary text-aware capability of SAM3 is transformed into stable class-level pixel responses, which provide the basis for subsequent global prior construction and dynamic fusion.

C. Divergence-Aware Global–Local Fusion

To alleviate the problems of incomplete local semantics, cross-window prediction inconsistency, and severe category ambiguity in remote sensing images, we propose a DGLF strategy. The key idea is to first construct a class-level global prior from the full-image perspective to provide complete scene-level semantic context, and then adaptively adjust the fusion strength between the local prediction and the global prior according to their distribution discrepancy. As illustrated in Fig. 3, the DGLF module improves the consistency of the category distribution within the target region and enhances the quality of local predictions.

Specifically, given an input image I, we first resize it to $\tilde{I}$ according to a predefined long-side resolution, and then apply the same open-vocabulary prediction process described in Section III-B to obtain the class-level prediction $\tilde{\mathbf{L}}$ at the resized scale. After that, $\tilde{\mathbf{L}}$ is restored to the original image resolution by interpolation, producing the full-image global prior map

$$\mathbf{G} = \mathrm{Resize}\left(\tilde{\mathbf{L}}, H, W\right) \tag{12}$$

where $\mathbf{G} \in \mathbb{R}^{C\times H\times W}$. Compared with local class prediction, $\mathbf{G}$ is obtained from full-image inference and thus provides more complete global scene layout and class co-occurrence information.

For the nth local window, let $\mathbf{L}^{(n)}$ denote its class-level local prediction, and let $\mathbf{G}^{(n)}$ denote the corresponding cropped region from the global prior. To measure their semantic consistency at each pixel, we first perform pixel-wise L1 normalization on the local prediction and the global prior, yielding the local class distribution and the global class distribution

$$\mathbf{P}^{(n)}(x,y) = \frac{\mathbf{L}^{(n)}(x,y)}{\sum_{c=1}^{C}\mathbf{L}_c^{(n)}(x,y) + \varepsilon} \tag{13}$$

$$\mathbf{Q}^{(n)}(x,y) = \frac{\mathbf{G}^{(n)}(x,y)}{\sum_{c=1}^{C}\mathbf{G}_c^{(n)}(x,y) + \varepsilon} \tag{14}$$

where ε is a small constant for numerical stability.

Based on the above distributions, we adopt the Jensen–Shannon divergence to measure the discrepancy between the local prediction and the global prior. Let

$$\mathbf{M}^{(n)}(x,y) = \frac{1}{2}\left(\mathbf{P}^{(n)}(x,y) + \mathbf{Q}^{(n)}(x,y)\right) \tag{15}$$

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then the discrepancy at pixel (x, y) is defined as

$$D_{\mathrm{JS}}^{(n)}(x,y) = \frac{1}{2}\mathrm{KL}\left(\mathbf{P}^{(n)}(x,y) \,\|\, \mathbf{M}^{(n)}(x,y)\right) + \frac{1}{2}\mathrm{KL}\left(\mathbf{Q}^{(n)}(x,y) \,\|\, \mathbf{M}^{(n)}(x,y)\right). \tag{16}$$

A larger $D_{\mathrm{JS}}^{(n)}(x,y)$ indicates a stronger discrepancy between the local prediction and the global prior at the current pixel, while a smaller value indicates higher consistency.

To adaptively regulate the injection strength of the global prior according to the discrepancy, we further define the local maximum class confidence as

$$C_{\mathrm{loc}}^{(n)}(x,y) = \max_c \mathbf{P}_c^{(n)}(x,y) \tag{17}$$

and construct the pixel-wise dynamic fusion weight as

$$W^{(n)}(x,y) = \alpha\left(\lambda_d\left(\frac{D_{\mathrm{JS}}^{(n)}(x,y)}{\log 2}\right)^{\gamma} + \lambda_c\left(1 - C_{\mathrm{loc}}^{(n)}(x,y)\right)\right) \tag{18}$$

where α is the base fusion coefficient, λ_d and λ_c control the contributions of the divergence term and the confidence term, respectively, and γ adjusts the sensitivity to the divergence term. The fused local–global prediction is then defined as

$$\hat{\mathbf{L}}^{(n)}(x,y) = \left(1 - W^{(n)}(x,y)\right)\mathbf{L}^{(n)}(x,y) + W^{(n)}(x,y)\,\mathbf{G}^{(n)}(x,y). \tag{19}$$

This fusion strategy enables the model to preserve local prediction stability when the local and global semantics are consistent, while adaptively strengthening the guidance of the global prior when the local prediction is uncertain or when the local–global discrepancy is large, thereby improving both global semantic consistency and local prediction robustness in open-vocabulary remote sensing semantic segmentation.

For clarity, the overall procedure of the proposed DGLF module is summarized in Algorithm 1. The algorithm describes how the full-image global prior is constructed and how it is adaptively fused with the local prediction according to the local–global discrepancy and local confidence.

Algorithm 1 Divergence-Aware Global–Local Fusion

Input: Input image I, local class prediction $\mathbf{L}^{(n)}$, long-side resolution R
Output: Fused local–global prediction $\hat{\mathbf{L}}^{(n)}$
Resize the input image:
$\tilde{I} \leftarrow \mathrm{Resize}(I, R)$
Predict the resized image by the SAM3-based open-vocabulary branch:
$\tilde{\mathbf{L}} \leftarrow \mathrm{OVPred}(\tilde{I})$
Construct the full-image global prior:
$\mathbf{G} \leftarrow \mathrm{Resize}(\tilde{\mathbf{L}}, H, W)$
Crop the prior patch for the current local window:
$\mathbf{G}^{(n)} \leftarrow \mathrm{Crop}(\mathbf{G}, n)$
Normalize the local and global predictions into class distributions:
$\mathbf{P}^{(n)} \leftarrow \mathrm{Norm}(\mathbf{L}^{(n)}), \quad \mathbf{Q}^{(n)} \leftarrow \mathrm{Norm}(\mathbf{G}^{(n)})$
Compute the mean distribution and discrepancy:
$\mathbf{M}^{(n)} \leftarrow \frac{1}{2}(\mathbf{P}^{(n)} + \mathbf{Q}^{(n)})$
$D_{\mathrm{JS}}^{(n)} \leftarrow \mathrm{JS}(\mathbf{P}^{(n)}, \mathbf{Q}^{(n)})$
Estimate the local confidence:
$C_{\mathrm{loc}}^{(n)} \leftarrow \max_c \mathbf{P}_c^{(n)}$
Generate the dynamic fusion weight:
$W^{(n)} \leftarrow \alpha\left[\lambda_d\left(\frac{D_{\mathrm{JS}}^{(n)}}{\log 2}\right)^{\gamma} + \lambda_c\left(1 - C_{\mathrm{loc}}^{(n)}\right)\right]$
Fuse the local prediction and the global prior:
$\hat{\mathbf{L}}^{(n)} \leftarrow (1 - W^{(n)})\mathbf{L}^{(n)} + W^{(n)}\mathbf{G}^{(n)}$
return $\hat{\mathbf{L}}^{(n)}$

D. Reliability-Gated Prediction Refinement

Although the divergence-aware fusion in Section III-C enables adaptive integration of local prediction and global prior, directly applying the fusion weight may still introduce unnecessary global correction in highly reliable local regions or propagate unreliable global priors into ambiguous areas. To address this issue, we further propose an RGPR strategy, which includes local trust suppression, ambiguous-region boosting, and global reliability gating. As illustrated in Fig. 4, the RGPR module corrects erroneously predicted regions and refines their boundaries, thereby improving prediction accuracy and reliability.

First, when the local prediction is already highly confident, the global prior should not dominate the fusion process. Therefore, we suppress prior injection in high-confidence local regions by setting

$$W^{(n)}(x,y) = 0, \quad \text{if } C_{\mathrm{loc}}^{(n)}(x,y) \geq \tau_h \tag{20}$$

where τ_h is the local trust threshold. This operation preserves sharp local structures and avoids oversmoothing caused by excessive global correction.

Second, for pixels whose local prediction is neither highly confident nor completely unreliable, the local–global discrepancy becomes more informative. We define such pixels as ambiguous regions and enhance the prior injection strength when the discrepancy is sufficiently large

$$W^{(n)}(x,y) \leftarrow \min\left(\beta W^{(n)}(x,y), 1\right) \tag{21}$$

if

$$\tau_l \leq C_{\mathrm{loc}}^{(n)}(x,y) \leq \tau_u \quad \text{and} \quad \frac{D_{\mathrm{JS}}^{(n)}(x,y)}{\log 2} \geq \tau_d \tag{22}$$

where τ_l and τ_u define the ambiguous confidence interval, τ_d is the discrepancy threshold, and $\beta > 1$ is the boost factor. In this way, the model more actively exploits the global prior in regions where local prediction is uncertain and inconsistent with the global context.

Third, the global prior itself may be unreliable in some regions. To prevent such unreliable priors from misleading the refinement process, we introduce a global reliability gate based on the maximum class confidence of the global prior

$$C_{\mathrm{glob}}^{(n)}(x,y) = \max_c \mathbf{Q}_c^{(n)}(x,y). \tag{23}$$

For the hard-gating form, the reliability map is defined as

$$R^{(n)}(x,y) = \begin{cases} 1, & C_{\mathrm{glob}}^{(n)}(x,y) \geq \tau_g \\ 0, & \text{otherwise} \end{cases} \tag{24}$$

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where τ_g is the global confidence threshold. The final gated fusion weight is then given by

$$\tilde{W}^{(n)}(x,y) = W^{(n)}(x,y)\, R^{(n)}(x,y). \tag{25}$$

Using the reliability-gated weight, the refined local–global prediction is obtained as

$$\tilde{\mathbf{L}}^{(n)}(x,y) = \left(1 - \tilde{W}^{(n)}(x,y)\right) \mathbf{L}^{(n)}(x,y) + \tilde{W}^{(n)}(x,y)\, \mathbf{G}^{(n)}(x,y). \tag{26}$$

After refinement, the predictions from all local windows are merged by overlap averaging to obtain the full-resolution fused response. Finally, the pixel-wise segmentation result is generated by

$$\hat{Y}(x,y) = \arg\max_c \tilde{\mathbf{L}}_c(x,y). \tag{27}$$

Overall, the proposed refinement strategy enforces the principle of trusting local prediction when it is reliable and exploiting global prior when it is necessary and trustworthy, thereby further improving boundary preservation, semantic consistency, and robustness in open-vocabulary remote sensing semantic segmentation.

For clarity, the overall procedure of the proposed reliability-gated refinement module is summarized in Algorithm 2. The algorithm shows how the initial fusion weight is adjusted through local trust suppression, ambiguity-aware boosting, and global reliability gating to produce the final refined prediction.

Algorithm 2 Reliability-Gated Prediction Refinement

Input: Local prediction $\mathbf{L}^{(n)}$, global prior $\mathbf{G}^{(n)}$, initial weight $W^{(n)}$
Discrepancy map $D_{\mathrm{JS}}^{(n)}$, thresholds $\tau_h, \tau_l, \tau_u, \tau_d, \tau_g$, boost factor β
Output: Refined prediction $\tilde{\mathbf{L}}^{(n)}$
Compute the normalized local and global distributions:
$\mathbf{P}^{(n)} \leftarrow \mathrm{Norm}(\mathbf{L}^{(n)}), \quad \mathbf{Q}^{(n)} \leftarrow \mathrm{Norm}(\mathbf{G}^{(n)})$
Estimate local and global confidence:
$C_{\mathrm{loc}}^{(n)} \leftarrow \max_c \mathbf{P}_c^{(n)}, \quad C_{\mathrm{glob}}^{(n)} \leftarrow \max_c \mathbf{Q}_c^{(n)}$
Suppress prior injection in highly reliable local regions:
for each pixel (x,y) **do**
  if $C_{\mathrm{loc}}^{(n)}(x,y) \geq \tau_h$ **then**
    $W^{(n)}(x,y) \leftarrow 0$
  end if
end for
Boost prior injection in ambiguous regions:
for each pixel (x,y) **do**
  if $\tau_l \leq C_{\mathrm{loc}}^{(n)}(x,y) \leq \tau_u$ **and** $\dfrac{D_{\mathrm{JS}}^{(n)}(x,y)}{\log 2} \geq \tau_d$ **then**
    $W^{(n)}(x,y) \leftarrow \min(\beta W^{(n)}(x,y), 1)$
  end if
end for
Construct the reliability gate:
for each pixel (x,y) **do**
  if $C_{\mathrm{glob}}^{(n)}(x,y) \geq \tau_g$ **then**
    $R^{(n)}(x,y) \leftarrow 1$
  else
    $R^{(n)}(x,y) \leftarrow 0$
  end if
end for
Obtain the gated fusion weight:
$\tilde{W}^{(n)} \leftarrow W^{(n)} \odot R^{(n)}$
Refine the local–global prediction:
$\tilde{\mathbf{L}}^{(n)} \leftarrow (1 - \tilde{W}^{(n)})\mathbf{L}^{(n)} + \tilde{W}^{(n)}\mathbf{G}^{(n)}$
return $\tilde{\mathbf{L}}^{(n)}$

IV. Experiment

In this section, extensive experiments are conducted to validate the effectiveness of the proposed method. We first describe the datasets and implementation details, followed by quantitative and qualitative comparisons with state-of-the-art (SOTA) methods. Furthermore, t-SNE visualization, computational cost analysis, and ablation studies are provided to comprehensively evaluate the discriminability, efficiency, robustness, and componentwise effectiveness of the proposed framework.

A. Dataset

To comprehensively evaluate the effectiveness and generalization ability of the proposed method for open-vocabulary remote sensing semantic segmentation, we conduct experiments on several representative remote sensing semantic segmentation datasets, including OpenEarthMap [49], LoveDA [50], iSAID [51], Potsdam,[1] Vaihingen,[2] UAVid [52], UDD5 [53], and VDD [54]. These datasets cover satellite, aerial, and UAV scenarios, with diverse spatial resolutions, scene complexities, and category definitions, thereby enabling a comprehensive evaluation of segmentation performance under different remote sensing settings. Among them, the first five datasets mainly consist of satellite images, while the last three are mainly composed of UAV images. Unless otherwise specified, we follow the official data splits or the experimental protocols adopted in previous studies for all datasets and use mIoU as the primary evaluation metric. More detailed implementation details, open-vocabulary settings, and comparison protocols are provided in Section IV-B.

B. Setup

To ensure fairness and reproducibility, all experiments are conducted on an NVIDIA RTX 4090 GPU and implemented based on the MM-Segmentation framework. Unless otherwise specified, the proposed method adopts a unified inference pipeline and parameter setting across all datasets.

Under the open-vocabulary setting, we construct a text prompt set for each semantic category based on the official category names of each dataset. When a category has common synonyms or ambiguous textual expressions, we include equivalent descriptions and aggregate their responses into the corresponding class score. For fair comparison, all compared methods are evaluated under the same category set, data split, label definition, and test protocol as SegEarth-OV3.

[1] https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam

[2] https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen

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TABLE I
DEFINITIONS OF EVALUATION METRICS USED IN THIS ARTICLE

Metric	Formula
mIoU	$\mathrm{mIoU} = \frac{1}{C}\sum_{c=1}^{C}\frac{TP_c}{TP_c + FP_c + FN_c}$
mAcc	$\mathrm{mAcc} = \frac{1}{C}\sum_{c=1}^{C}\frac{TP_c}{TP_c + FN_c}$
aAcc	$\mathrm{aAcc} = \frac{\sum_{c=1}^{C} TP_c}{\sum_{c=1}^{C}(TP_c + FN_c)}$
P90	$\mathrm{P90} = \mathrm{Percentile}_{90}(\{t_i\}_{i=1}^{N})$
PGM	$\mathrm{PGM} = \max_{1\le i\le N} m_i$

Note: mIoU denotes the mean Intersection over Union. mAcc denotes the mean class-wise pixel accuracy, and aAcc denotes the overall pixel accuracy over all valid pixels. P90 denotes the 90th-percentile inference latency over the test set, and PGM denotes Peak GPU Memory. C denotes the number of semantic categories. TP_c, FP_c, and FN_c denote the numbers of true-positive, false-positive, and false-negative pixels for class c, respectively. N denotes the number of test images. t_i and m_i denote the inference time and peak GPU memory usage of the i-th test image, respectively.

For reproducibility, the prompt lists are fixed for each dataset. All experiments use the same SAM3 image model, public checkpoint, BPE vocabulary file, and processor settings. The sliding-window configuration, global-prior resizing strategy, response aggregation rule, and threshold parameters are also kept fixed across datasets unless otherwise specified.

During inference, SAM3 is adopted as the basic open-vocabulary predictor, and sliding-window inference is used to obtain the initial class response maps. For the global-prior branch, the input image is resized to one half of its original long side while preserving the original aspect ratio, and the full-image global prior is constructed using the same open-vocabulary prediction process. For the local-prediction branch, the same sliding-window setting as SegEarth-OV3 is adopted. In the fusion stage, the key hyperparameters of the DGLF module are set as $\alpha = 0.25$, $\lambda_d = 0.7$, and $\lambda_c = 0.3$, while those of the reliability-gated refinement module are set as $\tau_h = 0.8$ and $\tau_l = 0.25$. All remaining hyperparameters are kept the same as those in SegEarth-OV3.

The above hyperparameters are selected by jointly considering their methodological roles and empirical stability across multiple datasets. Specifically, they are intended to balance the injection of full-image global priors and the preservation of reliable local predictions. The detailed selection rationale and cross-dataset sensitivity analysis are provided in the parameter ablation study in Section IV-H.

We compare the proposed method with several representative remote sensing open-vocabulary segmentation methods as well as general open-vocabulary segmentation methods. All compared methods follow their public implementations or the default settings reported in the corresponding papers as closely as possible. Unless otherwise specified, mIoU is used as the primary evaluation metric for semantic segmentation. To provide a more comprehensive evaluation, we additionally report mAcc and aAcc, which measure class-wise recognition accuracy and overall pixel-level accuracy, respectively. For computational cost analysis, P90 latency and Peak GPU Memory (PGM) are reported to evaluate inference stability and memory consumption. The definitions of these metrics are summarized in Table I.

C. Comparison to SOTA

Since the proposed GLoP-SAM3 is a training-free method and there are still relatively few OVSS methods specifically designed for remote sensing images, we select two representative OVSS methods for remote sensing imagery and two representative OVSS methods for natural images as baselines, namely SegEarth-OV, SegEarth-OV3, GEM, and ClearCLIP. In addition, to provide a more comprehensive qualitative comparison with recent open-vocabulary segmentation methods, we further include ProxyCLIP and CorrCLIP in the visualization analysis. As shown in Fig. 5, the updated qualitative comparison covers GEM, ClearCLIP, SegEarth-OV, SegEarth-OV3, ProxyCLIP, CorrCLIP, and the proposed GLoP-SAM3 on representative remote sensing scenes.

1) Overall Analysis: As shown in Table II, GLoP-SAM3 achieves the best mIoU performance among training-free methods on 7 out of the 8 semantic segmentation datasets. In particular, it improves the performance on OpenEarthMap by 3.7% and surpasses the current SOTA method by 0.8% in terms of average mIoU over the 8 datasets. Since GLoP-SAM3 is a training-free postprocessing framework built upon SAM3, these results indicate that the proposed global–local prior fusion and reliability-gated refinement mechanisms can effectively enhance the baseline model without introducing additional training.

Although mIoU is the primary metric for semantic segmentation, it mainly measures the average region-level overlap between predictions and ground-truth annotations and does not fully reflect class-level recognition accuracy or overall pixel-level prediction quality. Therefore, we further introduce mAcc and aAcc for a more comprehensive evaluation. Specifically, mAcc measures the mean class-wise pixel accuracy and reflects class-level recognition balance, while aAcc measures the overall accuracy over all valid pixels and reflects global pixel-level prediction quality.

As shown in Fig. 6, GLoP-SAM3 achieves positive improvements on most datasets and under most evaluation metrics. For example, on OpenEarthMap, GLoP-SAM3 improves mIoU, mAcc, and aAcc by +3.70, +3.60, and +1.50, respectively. On UDD5 and VDD, all three metrics also show consistent improvements. These results demonstrate that the proposed method improves not only region-level overlap quality but also class-wise recognition accuracy and overall pixel-level prediction quality in most remote sensing scenarios.

The numerical results in Table III further confirm the above observations. On OpenEarthMap, GLoP-SAM3 improves mAcc and aAcc by +3.6 and +1.5, respectively. On UAVid, mAcc improves by +1.4 while aAcc remains unchanged. On UDD5 and VDD, both mAcc and aAcc also increase. These results indicate that the improvement of GLoP-SAM3 is not

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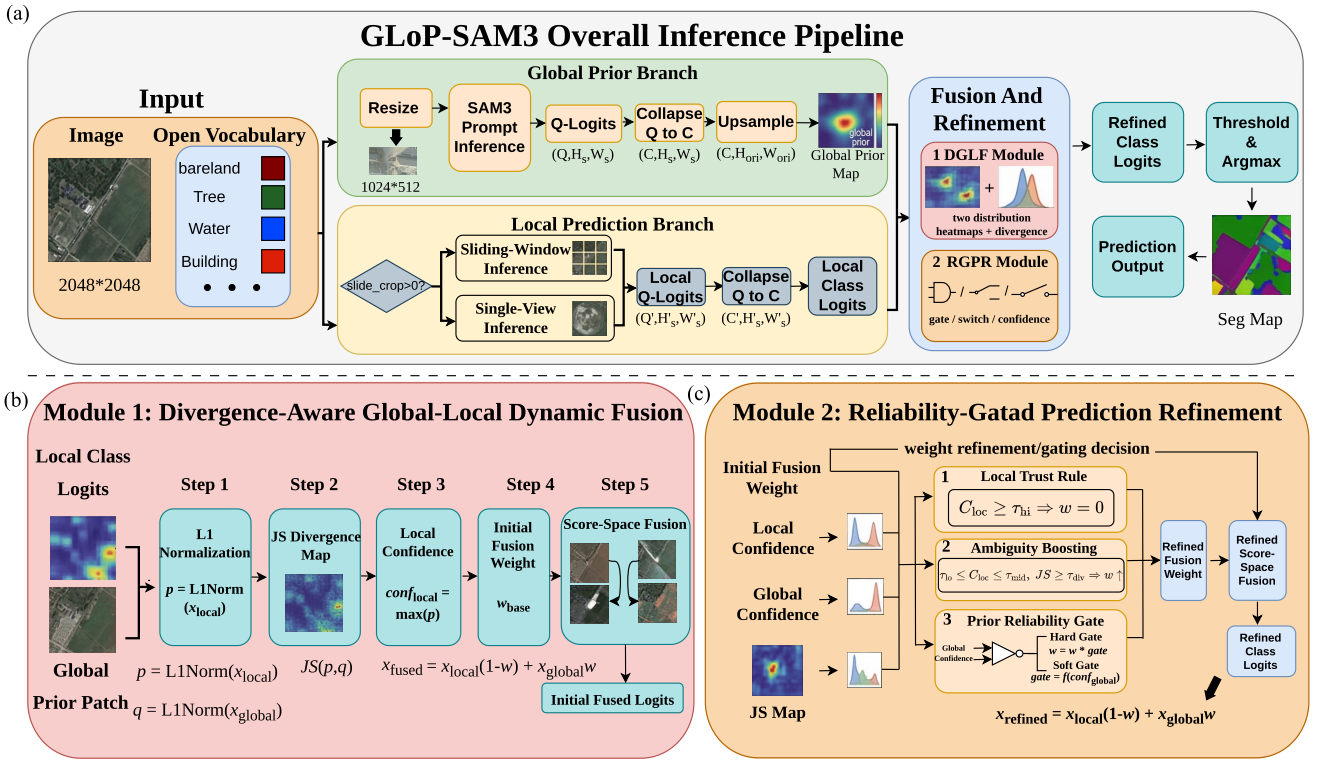


Fig. 2. Overview of the proposed GLoP-SAM3 framework and its core modules. (a) Overall inference pipeline of the proposed method, including the global prior branch, the local prediction branch, and the subsequent fusion and refinement process. (b) DGLF module, which generates dynamic fusion weights based on local–global distribution discrepancy and local confidence. (c) RGPR module, which further refines the fusion weights using local–global confidence and discrepancy information to improve prediction stability and boundary quality.

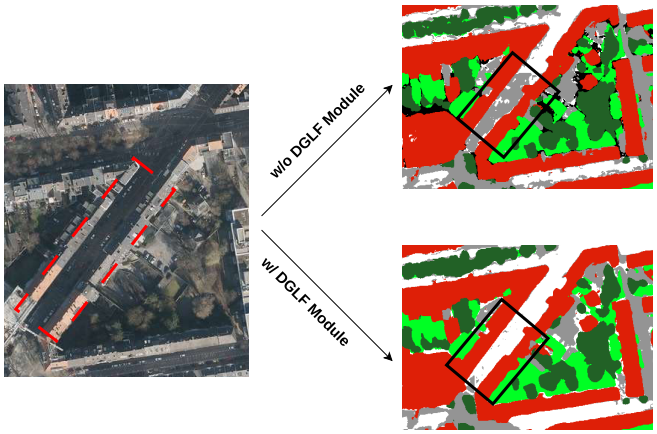


Fig. 3. Visualization example of the DGLF module. (Left panel) Input remote sensing image and the target region. (Upper-right and lower-right panels) Prediction results without and with the DGLF module, respectively. It can be observed that, after introducing DGLF, the category distribution within the target region becomes more consistent and the local prediction is noticeably improved.

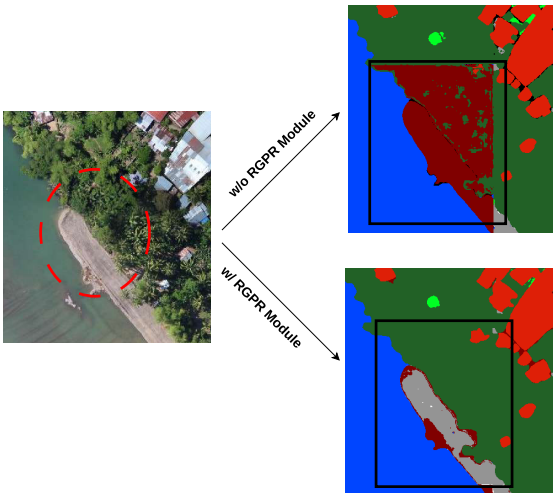


Fig. 4. Visualization example of the RGPR module. (Left panel) Input remote sensing image and the target region. (Upper-right and lower-right panels) Prediction results without and with the RGPR module, respectively. It can be observed that, after introducing RGPR, the target region is refined more accurately, and the boundary quality and prediction reliability are further improved.

limited to mIoU but is also reflected in additional evaluation dimensions.

Meanwhile, the multimetric results also provide a more objective view of dataset-specific limitations. For example, on Vaihingen, GLoP-SAM3 improves mIoU and aAcc by +1.40 and +2.90, respectively, but mAcc decreases by −1.20. This is likely because mAcc assigns equal weight to each class and is therefore more sensitive to recognition fluctuations in small or low-frequency categories, whereas aAcc is more influenced by the overall pixel accuracy of large-area classes. For LoveDA, although mAcc improves by +2.8, aAcc decreases by −3.8, suggesting that the class-wise average recognition ability is

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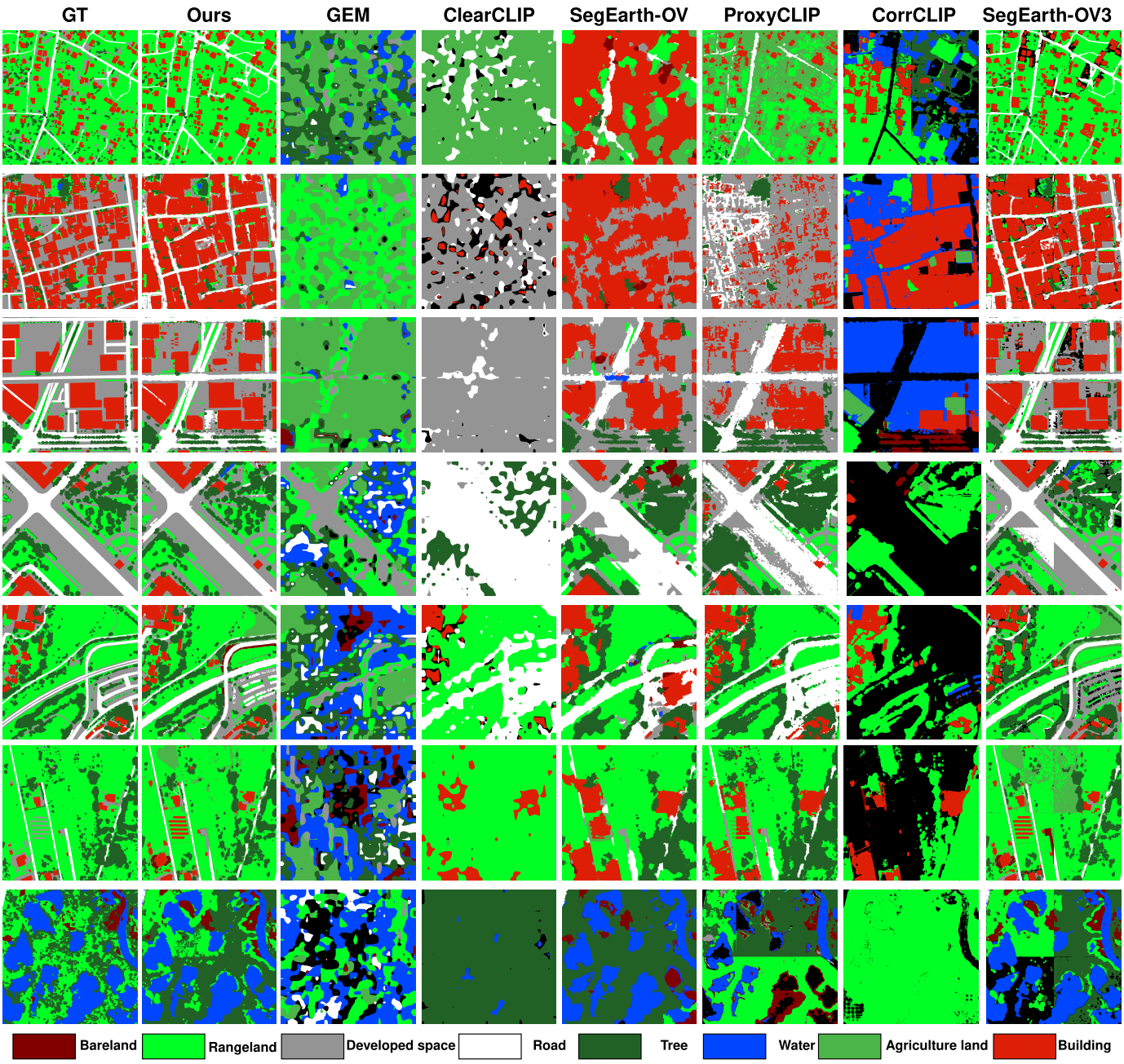


Fig. 5. Qualitative comparison of different open-vocabulary segmentation methods on representative remote sensing scenes. The compared methods include GEM [47], ClearCLIP [33], SegEarth-OV [9], SegEarth-OV3 [10], ProxyCLIP [48], CorrCLIP [35], and the proposed GLoP-SAM3. Compared with recent SOTA methods, GLoP-SAM3 produces more coherent segmentation results in complex remote sensing scenes, shows better global semantic consistency over large-scale land-cover regions, and reduces locally fragmented predictions. These results further demonstrate the effectiveness of the proposed global–local prior fusion and reliability-gated refinement strategy.

improved, while the overall pixel accuracy is affected by prediction changes in background or large-area categories. This observation is consistent with the background dominance and class-distribution imbalance discussed in the following LoveDA failure analysis.

Overall, the additional mAcc and aAcc results strengthen the experimental analysis by extending the evaluation from a single mIoU metric to a multimetric comparison. The radar visualization provides an intuitive overview of performance trends, while the numerical table gives detailed metric values. Together, these results provide a more comprehensive characterization of the effectiveness and limitations of GLoP-SAM3 in open-vocabulary remote sensing semantic segmentation.

2) Failure Analysis on LoveDA: Although GLoP-SAM3 improves performance on most datasets, it does not bring stable gains on LoveDA. As shown in Fig. 7, the failure cases on LoveDA mainly arise from two typical situations: background-dominated scenes and weakly annotated ambiguous regions.

First, in Case A, large background or invalid regions dominate both local predictions and full-image global priors. Since the proposed fusion process relies on normalized class response distributions, the relative responses

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TABLE II

OVSS QUANTITATIVE COMPARISON ON REMOTE SENSING DATASETS. EVALUATION METRIC: MIOU. **BEST** AND **SECOND-BEST** PERFORMANCES ARE HIGHLIGHTED. SCAN, SAN, SED, CAT-SEG, OVRS, GSNET, AND RSKT-SEG ARE TUNED ON DATASET (7002 IMAGES WITH 17 CATEGORIES). SKYSENSE-O IS TUNED ON SKY-SA DATASET (35 000 IMAGES WITH 1763 CATEGORIES). "ORACLE" IS ACHIEVED BY A FULLY SUPERVISED SEGFORMER-B0 MODEL USING FULL TRAINING DATA

Methods	OpenEarthMap	LoveDA	iSAID	Potsdam	Vaihingen	UAVid^{img}	UDD5	VDD	Avg
Training on remote sensing segmentation data									
SCAN [31]	-	23.2	44.3	27.5	15.2	20.3	34.1	29.2	-
SAN [24]	-	25.3	49.6	37.3	39.2	23.5	37.2	35.8	-
SED [25]	-	24.6	51.2	29.4	39.0	21.3	35.7	32.5	-
Cat-Seg [30]	-	28.6	53.3	35.8	42.3	25.7	40.2	39.1	-
OVRS [43]	-	31.5	52.7	36.4	43.5	24.1	40.8	37.2	-
GSNet [12]	-	32.5	**53.7**	37.9	44.1	24.2	40.9	37.3	-
RSKT-Seg [44]	-	33.2	**54.3**	38.4	42.7	25.7	42.1	39.7	-
SkySense-O [11]	40.8	38.3	43.9	54.1	51.6	-	-	-	-
Training-free									
CLIP [45]	12.0	12.4	7.5	15.6	10.8	10.9	9.5	14.2	11.4
MaskCLIP [28]	25.1	27.8	14.5	33.9	29.9	28.6	32.4	32.9	27.2
SCLIP [46]	29.3	30.4	16.1	39.6	35.9	31.4	38.7	37.9	31.1
GEM [47]	33.9	31.6	17.7	39.1	36.4	33.4	41.2	39.5	32.3
ClearCLIP [33]	31.0	32.4	18.2	42.0	36.2	36.2	41.8	39.3	33.4
SegEarth-OV [9]	40.3	36.9	21.7	48.5	40.0	42.5	50.6	45.3	39.2
ProxyCLIP [48]	38.9	34.3	21.8	49.0	47.5	35.8	40.8	47.8	39.5
CorrCLIP [35]	32.9	36.9	25.5	51.9	47.0	38.3	46.1	47.3	40.7
SegEarth-OV3 [10]	**42.9**	**47.4**	27.6	**57.8**	**60.8**	**54.7**	**71.7**	**64.5**	**53.4**
GLoP-SAM3(Ours)	**46.6**	**46.2**	28.0	**57.9**	**62.2**	**55.4**	**72.6**	**64.7**	**54.2**
Oracle	64.4	50.0	36.2	74.3	61.2	59.7	56.5	62.9	58.2

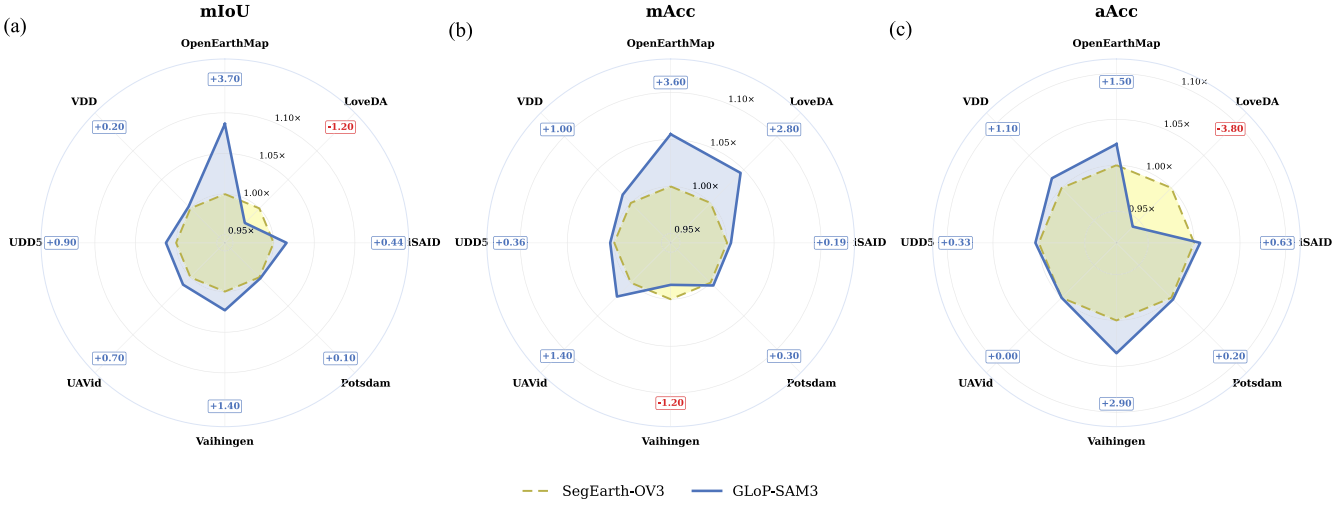


Fig. 6. Multimetric comparison between SegEarth-OV3 and GLoP-SAM3 on eight datasets. The three radar plots report (a) mIoU, (b) mAcc, and (c) aAcc, respectively. Each axis is normalized by the corresponding SegEarth-OV3 result, and the annotated values indicate the absolute gain of GLoP-SAM3 over SegEarth-OV3.

of true foreground objects, especially small or weak targets, can be suppressed when background responses occupy a large proportion of the image. As a result, the local–global discrepancy estimation and local confidence measurement become less reliable, causing valid foreground structures to be overwhelmed during the fusion process.

Second, in Case B, LoveDA contains weak boundaries, unlabeled regions, and ambiguous transitions between neighboring land-cover categories. When explicit background constraints are insufficient, the global prior may assign these uncertain regions to adjacent categories with stronger semantic responses. The divergence-aware fusion module may further propagate such responses into ambiguous areas, while the

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TABLE III
COMPARISON OF MACC AND AACC BETWEEN SEGEARTH-OV3 AND GLOP-SAM3

Dataset	Baseline mAcc	Ours mAcc	ΔmAcc	Baseline aAcc	Ours aAcc	ΔaAcc
OpenEarthMap [49]	64.8	68.4	+3.6	65.1	66.6	+1.5
LoveDA [50]	62.0	64.8	+2.8	63.8	60.0	-3.8
iSAID [51]	46.1	46.3	+0.2	95.8	96.4	+0.6
Potsdam[1]	72.2	72.5	+0.3	77.2	77.4	+0.2
Vaihingen[2]	78.4	77.2	-1.2	81.2	84.1	+2.9
UAVid^{img} [52]	67.3	68.7	+1.4	74.8	74.8	+0.0
UDD5 [53]	83.1	83.5	+0.4	87.1	87.5	+0.4
VDD [54]	81.5	82.5	+1.0	73.7	74.8	+1.1

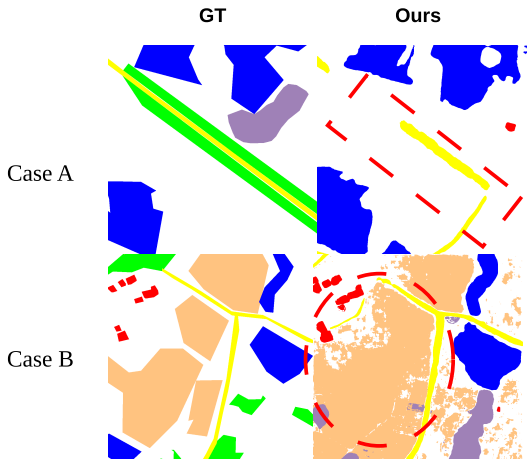


Fig. 7. Failure cases of GLoP-SAM3 on the LoveDA dataset. In Case A, dominant background or invalid regions suppress true foreground responses during the fusion process, causing target structures to be overwhelmed. In Case B, weakly annotated or unlabeled regions are incorrectly fused into adjacent categories with strong responses, leading to erroneous semantic propagation.

TABLE IV
PER-CLASS IOU ANALYSIS OF THE PERFORMANCE DEGRADATION ON LOVEDA

Class	SegEarth-OV3 IoU	GLoP-SAM3 IoU	ΔIoU
background	45.54	35.04	-10.50
building	63.80	64.27	+0.47
road	53.87	52.23	-1.64
water	51.44	57.03	+5.59
barren	35.84	35.72	-0.12
forest	33.85	33.97	+0.12
agricultural	47.50	44.86	-2.64
mIoU	47.41	46.16	-1.25

reliability-gated refinement module cannot always fully suppress this propagation when the global prior itself is unreliable. Consequently, unlabeled or inactive regions may be incorrectly fused into neighboring categories, leading to degraded segmentation performance.

Overall, these failure cases suggest that the performance drop on LoveDA is mainly related to background dominance, weak annotation boundaries, and uncertainty-driven error propagation, rather than a general failure of the proposed framework. This motivates the more detailed LoveDA-specific analyses below, including per-class IoU comparison, grouped error-propagation factor analysis, and class-level threshold control experiments.

To further verify the above analysis, we conduct three additional LoveDA-specific analyses, including per-class IoU comparison, grouped error-propagation factor analysis, and class-level threshold control experiments.

First, the per-class IoU results show that the performance degradation of GLoP-SAM3 on LoveDA does not occur uniformly across all classes. As shown in Table IV, the building, water, and forest classes still achieve positive IoU gains of +0.47, +5.59, and +0.12, respectively. However, the IoU of the background class decreases from 45.54 to 35.04, corresponding to a large drop of −10.50, which is one of the main factors contributing to the overall mIoU degradation on LoveDA. In addition, the road and agricultural classes decrease by −1.64 and −2.64, respectively. These results indicate that the degradation on LoveDA is mainly concentrated in background-related regions and several categories that are more prone to class confusion or spatial over-expansion, rather than being a uniform degradation across all semantic classes.

Second, we further analyze potential error-propagation factors from four perspectives: background ratio, void ratio, prediction disagreement, and dominant class ratio. As shown in Table V, images with a high background ratio obtain a Mean ΔmIoU of −2.73, whereas images with a low background ratio obtain a Mean ΔmIoU of only −0.47. This suggests that scenes dominated by background regions are more likely to suffer from stronger degradation. Meanwhile, images with high prediction disagreement obtain a Mean ΔmIoU of −2.20, compared with −1.00 for images with low prediction disagreement, indicating that error propagation becomes more severe when the predictions are more divergent. In addition, images with a high dominant class ratio obtain a Mean ΔmIoU of −1.97, while those with a low dominant class ratio obtain −1.23, suggesting that class-frequency imbalance and single-class dominance can further affect the reliability of the global prior. By contrast, the difference between high- and low-void-ratio groups is relatively small, indicating that the degradation on LoveDA cannot be simply attributed to void regions alone. Instead, it is more likely caused by the combined effect

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TABLE V
GROUPED STATISTICAL ANALYSIS OF ERROR-PROPAGATION FACTORS ON LOVEDA

Factor	Subset	Mean ΔmIoU	Interpretation	Implication
Void ratio	Low	-1.62	Void ratio alone is not the dominant factor.	Limited influence.
Void ratio	High	-1.47	Slightly smaller degradation.	Not the main cause.
Background ratio	Low	-0.47	Less affected.	Background-aware modeling is needed.
Background ratio	High	-2.73	Stronger degradation.	Major risk factor.
Prediction disagreement	Low	-1.00	More stable predictions.	Lower uncertainty.
Prediction disagreement	High	-2.20	Stronger error propagation.	Uncertainty calibration is beneficial.
Dominant class ratio	Low	-1.23	More balanced class composition.	Less class bias.
Dominant class ratio	High	-1.97	Stronger class-frequency imbalance.	Class-frequency handling is helpful.

TABLE VI
SUPPLEMENTARY VERIFICATION EXPERIMENTS OF CLASS-LEVEL THRESHOLD CONTROL ON LOVEDA

Class	Exp.1		Exp.2		Exp.3	
	Thr.	IoU	Thr.	IoU	Thr.	IoU
background	0.00	36.13	0.00	37.01	0.00	36.83
building	0.28	64.27	0.29	64.32	0.28	64.27
road	0.25	52.52	0.27	52.67	0.26	52.61
water	0.28	56.50	0.30	55.85	0.26	57.03
barren	0.27	35.84	0.28	35.86	0.26	35.72
forest	0.30	34.00	0.31	34.00	0.30	33.97
agricultural	0.29	45.10	0.31	45.37	0.31	45.37
mIoU	46.34 (+0.18)		46.44 (+0.28)		46.54 (+0.38)	

of background dominance, prediction uncertainty, and class-frequency imbalance.

Finally, to examine whether these factors can guide performance improvement, we further conduct class-level threshold control experiments. As shown in Table VI, without changing the SAM3 backbone, text prompts, or the overall inference pipeline, simply adjusting the prediction thresholds for different classes improves the mIoU on LoveDA from 46.16 of the original GLoP-SAM3 to 46.34, 46.44, and 46.54, respectively. Meanwhile, the background IoU improves from 35.04 to 37.01, the road IoU improves from 52.23 to 52.67, and the agricultural IoU improves from 44.86 to 45.37. These results indicate that lightweight class-level reliability control can partially alleviate the error propagation caused by background dominance and imbalanced class responses on LoveDA.

In summary, the additional analyses show that the degradation on LoveDA is mainly related to background dominance, prediction uncertainty, and class-frequency imbalance, rather than being caused by a single factor. The current lightweight reliability control in GLoP-SAM3 is still insufficient to fully resolve error propagation caused by complex backgrounds and class bias in LoveDA. Nevertheless, the supplementary experiments indicate that background-aware modeling, uncertainty calibration, and class-frequency-aware handling are effective and worthwhile directions for further improving the robustness of GLoP-SAM3 in complex remote sensing scenarios.

D. Statistical Significance Analysis

To further examine the statistical reliability of the performance differences between GLoP-SAM3 and SegEarth-OV3, we conduct an image-level paired statistical significance analysis. Specifically, each test image is treated as a paired sample. We compute the image-level mIoU of SegEarth-OV3 and GLoP-SAM3 on the same image, and define the paired difference as $\Delta\text{mIoU}_i = \text{mIoU}_i(\text{GLoP-SAM3}) - \text{mIoU}_i(\text{SegEarth-OV3})$. We then apply the Wilcoxon signed-rank test to all image-level ΔmIoU_i values and use paired bootstrap resampling to estimate the 95% confidence interval of the Mean ΔmIoU.

It should be noted that the Mean ΔmIoU in this statistical analysis is not exactly equivalent to the dataset-level mIoU difference reported in the main comparison table. The mIoU in the main table is a dataset-level metric computed after accumulating TP, FP, and FN for each class over the entire test set. In contrast, statistical significance testing requires a set of paired observations, and therefore we use image-level mIoU to construct the paired differences. This analysis is intended to evaluate whether the sample-level differences between the two methods are stable and should be regarded as a complement to, rather than a replacement for, the dataset-level mIoU comparison.

As shown in Table VII, GLoP-SAM3 obtains positive Mean ΔmIoU values on OpenEarthMap, iSAID, Potsdam, Vaihingen, UAVid, UDD5, and VDD. For most of these datasets, the p-values are below 0.05, indicating that the improvements exhibit favorable statistical stability. In particular, on OpenEarthMap, the Mean ΔmIoU reaches +2.80, with a 95% Bootstrap CI of [2.34, 3.23] and a Wilcoxon p-value of 9.07×10^{-32}, indicating a statistically significant improvement. Meanwhile, LoveDA obtains a Mean ΔmIoU of −3.04,

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TABLE VII
PAIRED STATISTICAL SIGNIFICANCE ANALYSIS RESULTS BETWEEN SEGEARTH-OV3 AND GLOP-SAM3

Dataset	Mean ΔmIoU	Median ΔmIoU	95% Bootstrap CI	Wilcoxon p-value	Direction	Significant
OpenEarthMap [49]	+2.80	+3.38	[2.34, 3.23]	9.07×10^{-32}	↑	True
LoveDA [50]	-3.04	-2.56	[-3.44, -2.62]	3.64×10^{-78}	↓	True
iSAID [51]	+0.44	0.00	[0.27, 0.61]	2.53×10^{-7}	↑	True
Potsdam[1]	+0.01	0.00	[0.002, 0.013]	4.61×10^{-2}	↑	True
Vaihingen[2]	+0.52	+0.75	[0.02, 1.01]	3.05×10^{-4}	↑	True
UAVid [52]	+0.46	+0.05	[0.26, 0.66]	1.34×10^{-5}	↑	True
UDD5 [53]	+0.76	+0.06	[0.35, 1.26]	1.60×10^{-4}	↑	True
VDD [54]	+0.65	+1.09	[0.14, 1.31]	2.19×10^{-4}	↑	True

Notes: ΔmIoU denotes the image-level mIoU difference between GLoP-SAM3 and SegEarth-OV3 on the same test image. Mean and Median ΔmIoU denote the average and median values over all paired test samples, respectively. The 95% Bootstrap CI is estimated by paired bootstrap resampling. The Wilcoxon p-value is obtained from the Wilcoxon signed-rank test. Significant=True means $p < 0.05$, indicating a statistically significant difference; when the Direction is ↓, it indicates a statistically significant degradation rather than an improvement.

with a 95% Bootstrap CI of [−3.44, −2.62] and a Wilcoxon p-value of 3.64×10^{-78}, showing that the degradation on LoveDA is also statistically significant.

Overall, the statistical analysis indicates that GLoP-SAM3 shows a stable positive trend on most remote sensing datasets, while LoveDA represents a clear limitation case. This result suggests that the average mIoU improvement should be interpreted more cautiously with respect to dataset-specific variations and also motivates a separate analysis of the performance degradation on LoveDA.

E. t-SNE Visualization

To further analyze the feature discriminability of different open-vocabulary segmentation methods, we visualize the final feature distributions of ProxyCLIP, CorrCLIP, SegEarth-OV, SegEarth-OV3, and the proposed GLoP-SAM3 using t-SNE on seven representative remote sensing datasets, including OpenEarthMap, iSAID, Potsdam, Vaihingen, UAVid, UDD5, and VDD. As shown in Fig. 8, this visualization provides an intuitive comparison of intraclass compactness and interclass separability across different methods.

From the t-SNE results, ProxyCLIP and CorrCLIP, as recent CLIP-based open-vocabulary segmentation methods, can form certain class-cluster structures on several datasets. However, their feature distributions still show relatively scattered intraclass patterns and noticeable overlaps among different categories. This indicates that directly relying on CLIP-based representations may still suffer from limited feature discrimination in complex remote sensing scenes, where different land-cover categories may exhibit similar textures, diverse scales, and ambiguous semantic boundaries. SegEarth-OV shows clearer clustering trends than the general CLIP-based methods on some remote sensing datasets, but interclass mixing and blurred cluster boundaries can still be observed in complex scenes. SegEarth-OV3 further improves the feature organization, yet some datasets still exhibit dispersed local clusters and overlaps between semantically adjacent categories.

In contrast, GLoP-SAM3 generally produces more compact intraclass aggregation and clearer interclass separation on most datasets. In particular, on OpenEarthMap, Potsdam, UAVid, UDD5, and VDD, the class clusters produced by GLoP-SAM3 are more concentrated, and the boundaries between different categories are more distinguishable. This suggests that the proposed global–local prior fusion and reliability-gated refinement not only improve the final segmentation predictions but also enhance the semantic organization of the feature space. Specifically, the global prior helps alleviate unstable local responses and interclass confusion caused by sliding-window inference, while the reliability-gated refinement suppresses unreliable prior propagation in ambiguous regions. As a result, the enhanced feature distributions show better intraclass consistency and interclass separability.

It should be noted that t-SNE visualization is mainly used for qualitative analysis, and the 2-D projection may be affected by the dimensionality reduction process itself. Therefore, it cannot replace quantitative evaluation. Nevertheless, together with the quantitative results and qualitative comparisons, the t-SNE results further indicate that GLoP-SAM3 can produce more discriminative feature distributions than ProxyCLIP, CorrCLIP, SegEarth-OV, and SegEarth-OV3 on most datasets, demonstrating its effectiveness in improving feature discriminability and semantic consistency for open-vocabulary remote sensing semantic segmentation.

F. Computational Cost Analysis

To further evaluate the deployment efficiency of the proposed method, we compare SegEarthOV3, GLoP-SAM3, and two CLIP+SAM3 variants from three complementary perspectives, namely segmentation accuracy, latency stability, and memory overhead. As shown in Fig. 9(a)–(c), GLoP-SAM3 achieves the highest mIoU on the OpenEarthMap dataset, while the two CLIP+SAM3 variants exhibit lower latency in some cases but still fail to surpass GLoP-SAM3 in overall segmentation accuracy. In particular, the ViT-L/14 variant introduces a higher GPU memory overhead. These observations suggest that directly attaching an additional pretrained model to SAM3 does not necessarily yield the best performance-cost trade-off.

From a theoretical perspective, let Q denote the number of text queries, T the number of local windows, C the number of semantic classes, and $H \times W$ the image resolution.

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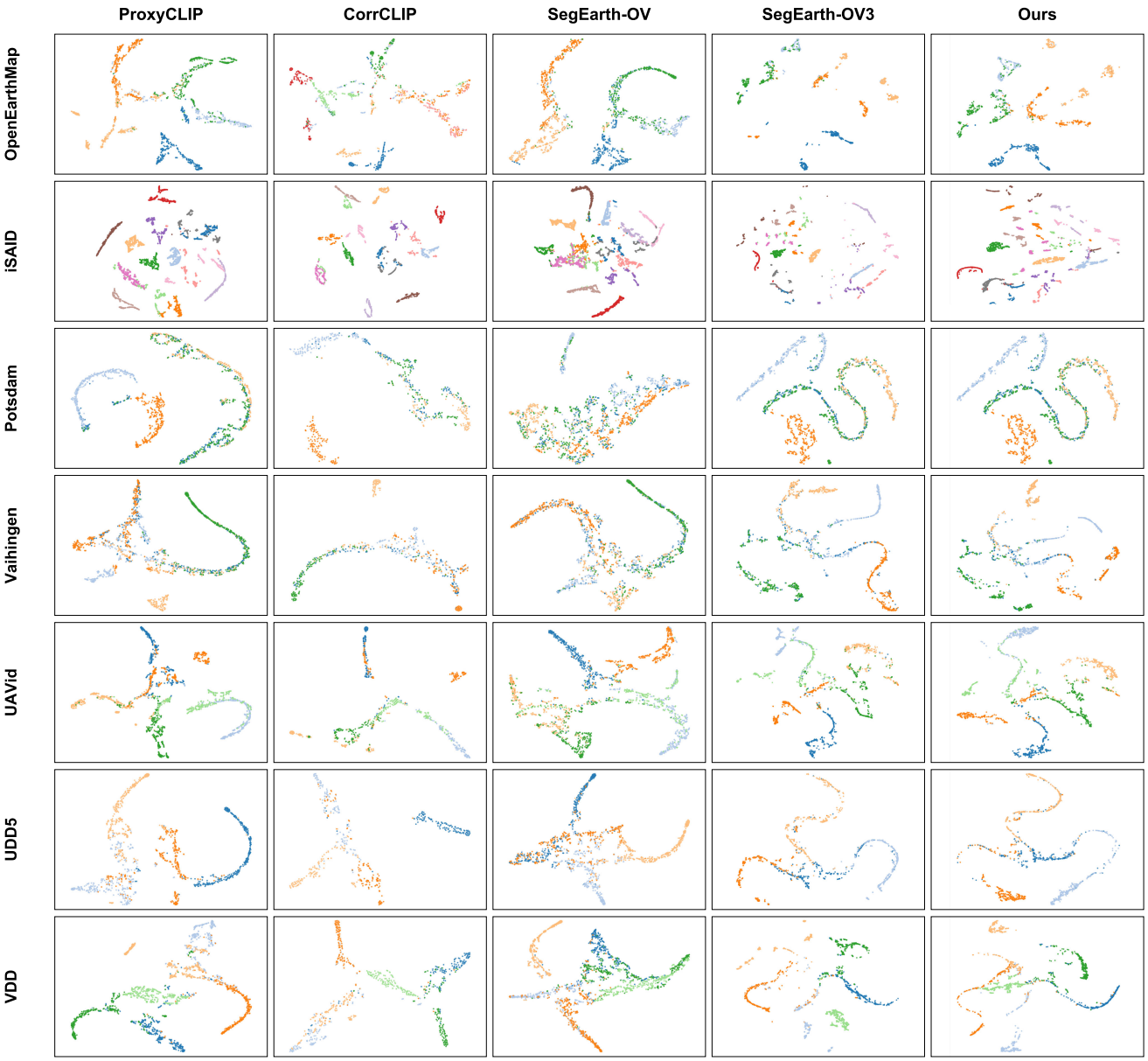


Fig. 8. t-SNE visualization comparison with recent open-vocabulary segmentation methods on multiple remote sensing datasets. The compared methods include ProxyCLIP [48], CorrCLIP [35], SegEarth-OV [9], SegEarth-OV3 [10], and the proposed GLoP-SAM3. The rows correspond to different datasets, including OpenEarthMap [49], iSAID [51], Potsdam[1], Vaihingen[2], UAVid [52], UDD5 [53], and VDD [54].

The complexity of SegEarthOV3 is dominated by repeated SAM3-based open-vocabulary inference over all windows and queries, which can be written as

$$\mathcal{O}\left(TQf_{\mathrm{SAM3}}(h,w)\right) \tag{28}$$

where $h \times w$ denotes the local window resolution. Compared with the baseline, GLoP-SAM3 additionally introduces one low-resolution global-prior branch and a lightweight pixel-wise fusion/refinement process, resulting in

$$\mathcal{O}\left(TQf_{\mathrm{SAM3}}(h,w) + Qf_{\mathrm{SAM3}}\left(H_g, W_g\right) + CHW\right) \tag{29}$$

where $H_g \times W_g$ denotes the resolution of the global-prior branch. Therefore, GLoP-SAM3 is indeed slightly more complex than SegEarthOV3, but its additional cost mainly comes from a single reduced-resolution prior inference and simple tensor-level fusion operations, rather than from introducing a new heavy backbone or extra trainable modules.

To further examine whether directly adding an external pretrained model on top of SAM3 can produce a better cost-effectiveness trade-off, we additionally construct two naive CLIP+SAM3 variants as reference methods. It should be emphasized that this comparison is not intended to establish a new SOTA baseline, but rather to verify that simply augmenting SAM3 with an extra pretrained vision-language backbone does not necessarily lead to the best overall performance–cost balance. Similar to our method, these CLIP+SAM3 variants are also fully training-free. In this setting, SAM3 is responsible for generating open-vocabulary segmentation responses and candidate regions, while CLIP is used to perform region-level semantic rescoring over the candidate regions. To ensure a fair

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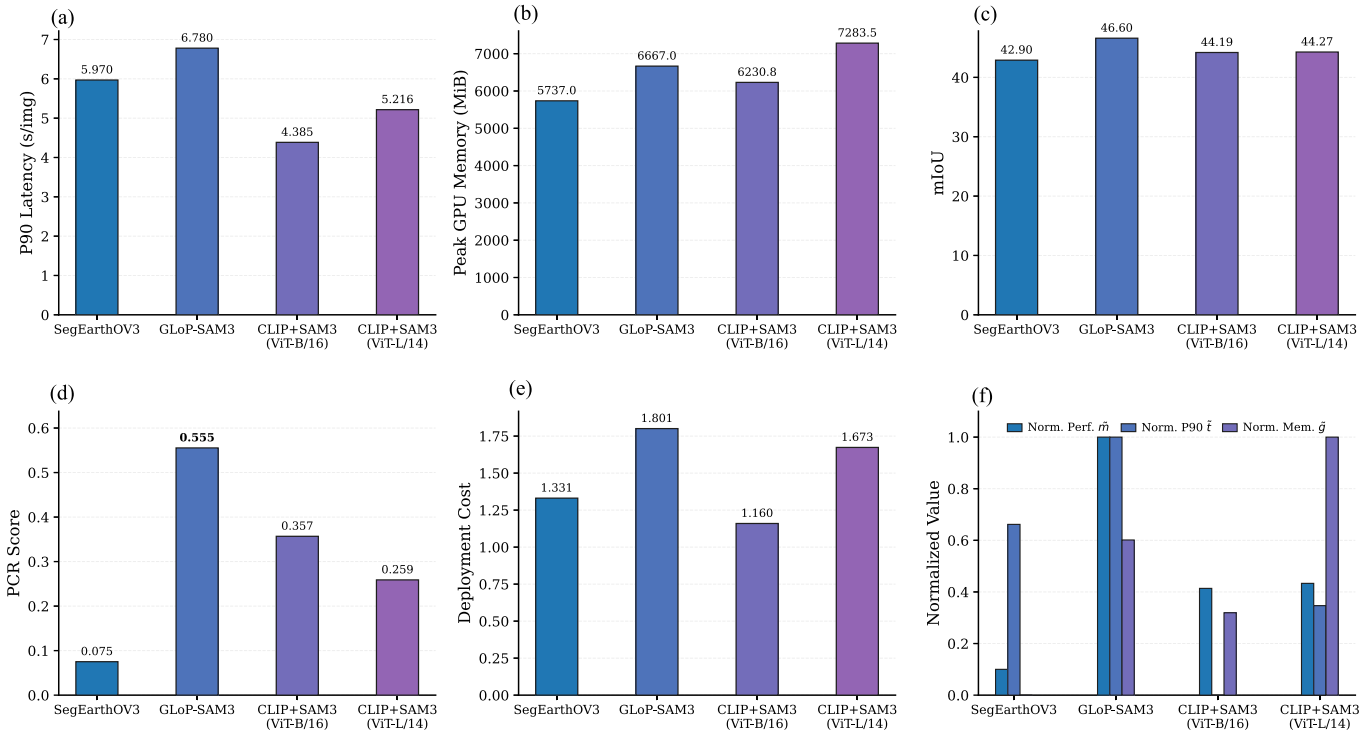


Fig. 9. Comparison of SegEarthOV3, GLoP-SAM3, and CLIP+SAM3 variants on the OpenEarthMap dataset. (a) P90 latency. (b) PGM. (c) mIoU. *(d)* PCR. (e) Normalized deployment cost in PCR. (f) Normalized performance and cost components used in PCR computation.

comparison, we tune the key CLIP-related hyperparameters and report the best CLIP+SAM3 results obtained on the OEM dataset for both ViT-B/16 and ViT-L/14.

Under a common candidate-region re-scoring formulation, the complexity of CLIP+SAM3 can be approximated as

$$\mathcal{O}\left(TQf_{\mathrm{SAM3}}(h,w) + Mf_{\mathrm{CLIP}}(u,v) + MCd\right) \tag{30}$$

where M denotes the number of candidate regions, $u \times v$ denotes the region resolution, and d denotes the embedding dimension for region–text similarity matching. Compared with GLoP-SAM3, CLIP+SAM3 requires an additional external vision-language model for region-level inference, and its extra complexity grows with the number of candidate regions. Therefore, it serves as a representative naive solution for evaluating whether adding an external pretrained backbone to SAM3 can produce a better trade-off in practice. As shown in Fig. 9(a)–(c), although CLIP+SAM3 may achieve lower latency in certain settings, its segmentation accuracy remains below that of GLoP-SAM3, indicating that naively introducing an external pretrained model does not provide the best cost-effectiveness.

To better summarize the performance–cost relationship, we further define a *Performance-Cost Ratio* (PCR). Unlike a simple weighted sum of accuracy and efficiency metrics, PCR explicitly adopts a benefit-over-cost formulation, where the numerator denotes normalized performance gain and the denominator denotes normalized deployment cost. Since OVSS primarily targets prediction quality, while P90 latency and PGM reflect practical deployment constraints, we retain three representative metrics only, namely mIoU, P90 latency, and PGM.

Let m_i, t_i, and g_i denote the mIoU, P90 latency, and PGM of the ith method, respectively. We first define a normalized performance term with a lower bound

$$\tilde{m}_i = \varepsilon + (1-\varepsilon)\frac{m_i - \min_j m_j}{\max_j m_j - \min_j m_j}$$

where ε is a small constant introduced to avoid degeneration of the numerator for the worst-performing method. For P90 latency and PGM, the normalized cost terms are defined as

$$\tilde{t}_i = \frac{t_i - \min_j t_j}{\max_j t_j - \min_j t_j}, \quad \tilde{g}_i = \frac{g_i - \min_j g_j}{\max_j g_j - \min_j g_j}.$$

Based on these terms, the normalized deployment cost is defined as

$$\mathrm{Cost}_i = 1 + \lambda_t \tilde{t}_i + \lambda_g \tilde{g}_i$$

where λ_t and λ_g are the weights for latency and memory cost, respectively, with $\lambda_t + \lambda_g = 1$. The final PCR is then defined as

$$\mathrm{PCR}_i = \frac{\tilde{m}_i}{1 + \lambda_t \tilde{t}_i + \lambda_g \tilde{g}_i}. \tag{31}$$

This formulation is adopted for three reasons. First, PCR follows an explicit performance-over-cost structure, which is more consistent with the intuitive meaning of cost-effectiveness than a direct linear combination. Second, it avoids repeatedly counting multiple highly correlated speed indicators, such as mean latency, P50 latency, P90 latency, and FPS, in a single score. Third, P90 latency is more informative than mean latency for characterizing tail-delay risk in practical deployment, while PGM directly measures resource consumption. Fig. 9(e) and (f) further visualize the

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resulting deployment cost and the normalized components used in PCR computation.

As shown in Fig. 9(d), GLoP-SAM3 achieves the highest PCR score among all compared methods. This result indicates that, although GLoP-SAM3 introduces a moderate increase in latency and memory relative to SegEarthOV3, its accuracy gain is substantial enough to compensate for this overhead, thereby yielding the best overall cost-effectiveness. In other words, the advantage of GLoP-SAM3 does not lie in pursuing the lowest deployment cost alone, but in achieving a more favorable trade-off: it obtains clearly stronger segmentation performance with only a controlled increase in deployment cost, without relying on any additional external pretrained model.

It should also be emphasized that GLoP-SAM3 is not intended to replace SegEarthOV3 as a universally low-latency inference mode. Its additional overhead mainly comes from the construction of the full-image global prior and the subsequent global–local fusion/refinement, rather than from an additional trainable backbone or an extra training stage. Therefore, the deployment value of GLoP-SAM3 should be interpreted under an accuracy-oriented offline mapping setting. In large-scale land-cover mapping, urban land-cover inventory, postdisaster interpretation, and open-category semantic analysis, inference is often performed in an offline or batch-processing manner, where semantic consistency and the reduction of erroneous regions are usually more important than the real-time latency of a single image. From this perspective, the average mIoU gain of GLoP-SAM3 is meaningful because it is obtained under the training-free and open-vocabulary setting without modifying the SAM3 backbone or introducing additional trainable models.

Nevertheless, for strict real-time processing, edge-device deployment, or memory-constrained applications, SegEarthOV3 remains a more lightweight choice. In practical use, GLoP-SAM3 can be regarded as an enhanced mode for high-accuracy offline mapping, while SegEarthOV3 can serve as the basic mode for efficiency-sensitive scenarios. The computational cost of GLoP-SAM3 is also configurable by reducing the long-side resolution of the global prior, simplifying sliding-window inference, tuning the prior fusion strength, or enabling the refinement only for low-confidence or highly uncertain regions.

Overall, these results demonstrate that GLoP-SAM3 provides a more effective balance among complexity, deployment overhead, and segmentation quality. Compared with SegEarthOV3, it achieves higher segmentation accuracy with an acceptable additional cost; compared with naive CLIP+SAM3 combinations, it offers better overall cost-effectiveness and stronger practical value for training-free open-vocabulary remote sensing semantic segmentation.

G. Parameter-Efficiency Analysis

In addition to segmentation accuracy and inference cost, parameter efficiency is an important aspect for evaluating the practical applicability of OVSS methods. In open-vocabulary remote sensing scenarios, a desirable method should not only generalize to text-defined categories but also avoid excessive task-specific training, additional learnable modules, and parameter-update overhead. Therefore, we further analyze representative OVSS methods from the perspectives of frozen parameters, extra trainable parameters, and average segmentation performance.

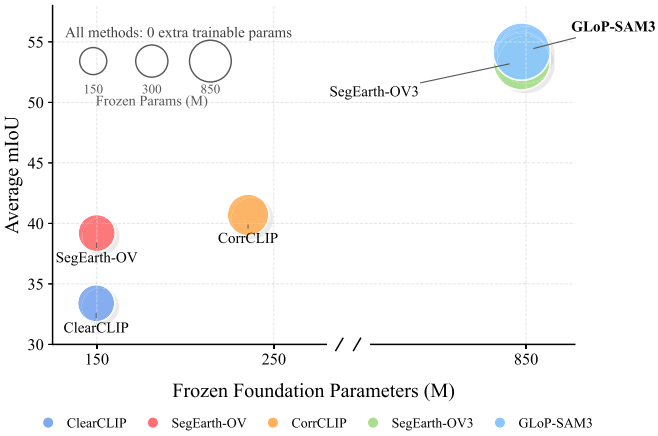


Fig. 10. Parameter-performance comparison among training-free OVSS methods. All compared methods introduce zero extra trainable parameters. The bubble size denotes the number of frozen foundation parameters.

Table VIII summarizes the parameter scale and performance of different types of methods. The selected methods cover training-based OVSS methods, training-based remote sensing OVSS methods, remote sensing foundation models, and training-free OVSS methods. Specifically, CAT-Seg, RSKT-Seg, and SkySense-O represent parameterized methods that require additional training or task-specific adaptation. ClearCLIP, CorrCLIP, SegEarth-OV, and SegEarth-OV3 represent training-free open-vocabulary segmentation methods. GLoP-SAM3 is the proposed training-free remote sensing open-vocabulary segmentation framework built upon frozen SAM3.

In this analysis, Frozen Params denote nontrainable pretrained parameters used during inference, such as the parameters of CLIP, SAM, or SAM3. Extra Trainable Params denote additional learnable parameters introduced beyond the adopted foundation model or backbone and updated during task-specific training or adaptation. For training-free methods, the model weights are not updated in a task-specific manner; therefore, their extra trainable parameters are zero.

As shown in Table VIII, training-based methods usually rely on additional trainable modules or task-specific adaptation. For example, CAT-Seg and RSKT-Seg introduce 4.67 and 29.89 M extra trainable parameters, respectively, while SkySense-O introduces 1253.15 M extra trainable parameters. Although these methods can obtain task-specific adaptation through training, they also require additional parameter updates, gradient storage, and optimizer states. In contrast, training-free methods avoid updating model weights, which provides better flexibility for open-category and cross-scene deployment.

Fig. 10 further visualizes the parameter-performance relationship among representative training-free methods. All

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TABLE VIII
PARAMETER-EFFICIENCY COMPARISON OF REPRESENTATIVE OVSS METHODS

Method	Type	Foundation Model / Backbone	Frozen Params	Extra Trainable Params	Avg. mIoU
		Training-based methods			
CAT-Seg [30]	General OVSS	CLIP + decoder	149.62M	4.67M	–
RSKT-Seg [44]	RS-OVSS	RS segmentation network	266.26M	29.89M	–
SkySense-O [11]	RS foundation model	SkySense	54.53M	1253.15M	–
		Training-free methods			
ClearCLIP [33]	General OVSS	CLIP	149.62M	0	33.4
CorrCLIP [35]	General OVSS	CLIP	235.43M	0	40.7
SegEarth-OV [9]	RS-OVSS	CLIP/SAM-based	149.91M	0	39.2
SegEarth-OV3 [10]	RS-OVSS	SAM3	848M	0	53.4
GLoP-SAM3(Ours)	**RS-OVSS**	**SAM3 + DGLF/RGPR**	**848M**	**0**	**54.2**

Notes: Frozen Params denote non-trainable pretrained parameters used during inference. Extra Trainable Params denote additional learnable parameters introduced beyond the adopted foundation model or backbone and updated during task-specific training or adaptation. "–" indicates that the 8-dataset average mIoU is not reported because the method is not evaluated on all eight datasets.

methods shown in the figure introduce zero extra trainable parameters, and the bubble size denotes the scale of frozen foundation parameters. It can be observed that CLIP-based methods, such as ClearCLIP and CorrCLIP, use relatively smaller frozen foundation models but achieve lower average mIoU values. SegEarth-OV further adapts open-vocabulary segmentation to remote sensing scenarios, but its average mIoU remains lower than that of SAM3-based methods.

It is worth noting that SAM3-based methods have a larger number of frozen parameters than CLIP-based methods. However, these parameters come from the frozen SAM3 foundation model and are not involved in task-specific training or parameter updates in our framework. Therefore, Frozen Params mainly reflect the scale of the adopted foundation model and should not be directly interpreted as additional training cost. The main focus of this analysis is not to simply compare the total number of frozen foundation-model parameters, but to examine whether a method introduces extra trainable parameters and whether it can more effectively exploit the outputs of an existing frozen model.

A direct comparison between SegEarth-OV3 and GLoP-SAM3 provides a clearer view of the proposed method. Both methods are built upon the same frozen SAM3 foundation model, have the same number of frozen parameters, and introduce zero extra trainable parameters. Nevertheless, GLoP-SAM3 improves the average mIoU from 53.4 to 54.2 through global–local prior fusion and reliability-gated refinement. This indicates that the performance gain of GLoP-SAM3 does not come from increased model capacity or additional training, but from a more effective organization, fusion, and reliability control of the output responses produced by frozen SAM3.

Overall, although SAM3-based methods use a larger frozen foundation model, GLoP-SAM3 achieves further improvement under the same frozen parameter scale as SegEarth-OV3, while maintaining zero extra trainable parameters. These results demonstrate that the proposed method improves the utilization efficiency of frozen SAM3 representations rather than relying on additional model capacity or task-specific training, thereby achieving a favorable balance among parameter efficiency, open-vocabulary generalization, and applicability to remote sensing scenarios.

H. Ablation Studies

1) Quantitative Analysis:

a) Module ablation: To systematically evaluate the contribution of each module to the model performance, we conduct module-level ablation experiments on multiple representative remote sensing datasets. Specifically, three experimental settings are considered: baseline denotes using SAM3 alone as the basic open-vocabulary predictor; +M1 denotes adding the DGLF module to the baseline; and +M1 +M2 further introduces the RGPR module, corresponding to the complete GLoP-SAM3 framework. To avoid relying only on a single mIoU metric, we report mIoU, mAcc, and aAcc simultaneously, so as to analyze the effects of different modules from the perspectives of region-level overlap quality, class-level recognition ability, and overall pixel-level prediction accuracy.

As shown in Table IX, with the progressive introduction of M1 and M2, the model shows a stable improvement trend in mIoU on most datasets. For example, on OpenEarthMap, the mIoU/mAcc/aAcc values of the baseline are 42.9/64.8/65.1. After introducing M1, the results improve to 45.5/67.5/66.0, and further increase to 46.6/68.4/66.6 after adding M2. This indicates that the two modules not only improve region-level overlap quality but also enhance class-level average accuracy and overall pixel-level accuracy. Similarly, on UDD5 and VDD, the complete model achieves improvements in all three metrics, demonstrating that DGLF and RGPR have favorable generalization contributions across different remote sensing scenarios.

Further observations show that M1 usually brings the main performance gain, suggesting that global–local prior fusion helps alleviate unstable local responses caused by

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TABLE IX
COMPLETE ABLATION EXPERIMENTS WITH ADDITIONAL MACC AND AACC METRICS

Dataset	Setting	SAM3	M1	M2	mIoU	mAcc	aAcc
OpenEarthMap [49]	Baseline	✓			42.9	64.8	65.1
	+M1	✓	✓		45.5	67.5	66.0
	+M1+M2	✓	✓	✓	46.6	68.4	66.6
iSAID [51]	Baseline	✓			27.6	46.1	95.8
	+M1	✓	✓		27.9	34.7	95.9
	+M1+M2	✓	✓	✓	28.0	46.3	96.4
Potsdam[1]	Baseline	✓			57.8	72.2	77.2
	+M1	✓	✓		57.85	72.4	75.7
	+M1+M2	✓	✓	✓	57.94	72.5	77.4
Vaihingen[2]	Baseline	✓			60.8	78.4	81.2
	+M1	✓	✓		61.9	78.2	79.4
	+M1+M2	✓	✓	✓	62.2	77.2	84.1
UAVid^{img} [52]	Baseline	✓			54.7	67.3	74.8
	+M1	✓	✓		55.0	68.5	74.8
	+M1+M2	✓	✓	✓	55.4	68.7	74.8
UDD5 [53]	Baseline	✓			71.7	83.1	87.1
	+M1	✓	✓		72.2	83.4	87.3
	+M1+M2	✓	✓	✓	72.6	83.5	87.5
VDD [54]	Baseline	✓			64.5	81.5	73.7
	+M1	✓	✓		64.6	82.3	74.2
	+M1+M2	✓	✓	✓	64.7	82.5	74.8

sliding-window inference and improves global semantic consistency. Based on M1, M2 further refines the prediction results through high-confidence region preservation, ambiguous-region enhancement, and global-prior reliability gating, thereby further improving the performance on datasets such as OpenEarthMap, UAVid, UDD5, and VDD. Meanwhile, slight fluctuations in mAcc or aAcc can be observed on some datasets. For instance, on Vaihingen, the complete model improves mIoU and aAcc, while mAcc decreases slightly. This indicates that different modules may have nonidentical effects on class-level recall and overall pixel prediction, which further demonstrates the necessity of reporting mIoU, mAcc, and aAcc simultaneously in the ablation study. Overall, the module ablation experiments verify that both M1 and M2 make practical contributions to the final performance and jointly enhance GLoP-SAM3 from the perspectives of global semantic consistency and local reliability refinement.

b) Parameter ablation: To analyze the influence of key hyperparameters on model performance, we conduct parameter ablation experiments on seven representative remote sensing datasets. Specifically, we investigate α, λ_d, and λ_c in DGLF, as well as τ_h and τ_l in RGPR. Here, α controls the injection strength of the global prior, λ_d and λ_c regulate the effects of local–global distribution discrepancy and local confidence, respectively, while τ_h and τ_l are used to control high-confidence region preservation and ambiguous-region enhancement.

As shown in Fig. 11(a)–(e), the mIoU curves on different datasets vary smoothly under different parameter values, indicating that GLoP-SAM3 is relatively robust to key hyperparameters. Specifically, $\alpha = 0.25$ achieves a favorable balance between global context utilization and local reliability preservation. $\lambda_d = 0.7$ and $\lambda_c = 0.3$ effectively combine local–global distribution discrepancy and local confidence information, thereby more accurately identifying regions that require refinement. In addition, $\tau_h = 0.8$ and $\tau_l = 0.25$ help preserve high-confidence local predictions while moderately enhancing ambiguous regions.

Furthermore, Fig. 11(f)–(h) reports the average mIoU, mAcc, and aAcc under five candidate settings. It can be observed that the default hyperparameter setting achieves the best or near-best performance across all three average metrics, and the performance variations among neighboring settings are relatively small. This demonstrates that the adopted hyperparameter configuration is not an empirical choice tailored to a single dataset, but exhibits favorable stability across multiple remote sensing datasets and evaluation metrics. Overall, the parameter ablation results verify that moderate global-prior fusion, distribution-discrepancy awareness, and reliability gating are effective for improving the robustness and generalization ability of GLoP-SAM3.

The final default setting, i.e., $\alpha = 0.25$, $\lambda_d = 0.7$, $\lambda_c = 0.3$, $\tau_h = 0.8$, and $\tau_l = 0.25$, is therefore determined according to both methodological considerations and experimental validation. From the methodological perspective, α should be large enough to introduce useful global context, but not too large to cause excessive propagation of unreliable global priors. λ_d and λ_c are selected to jointly emphasize local–global distribution discrepancy and local confidence, so that the global prior is mainly introduced in regions that are uncertain or inconsistent

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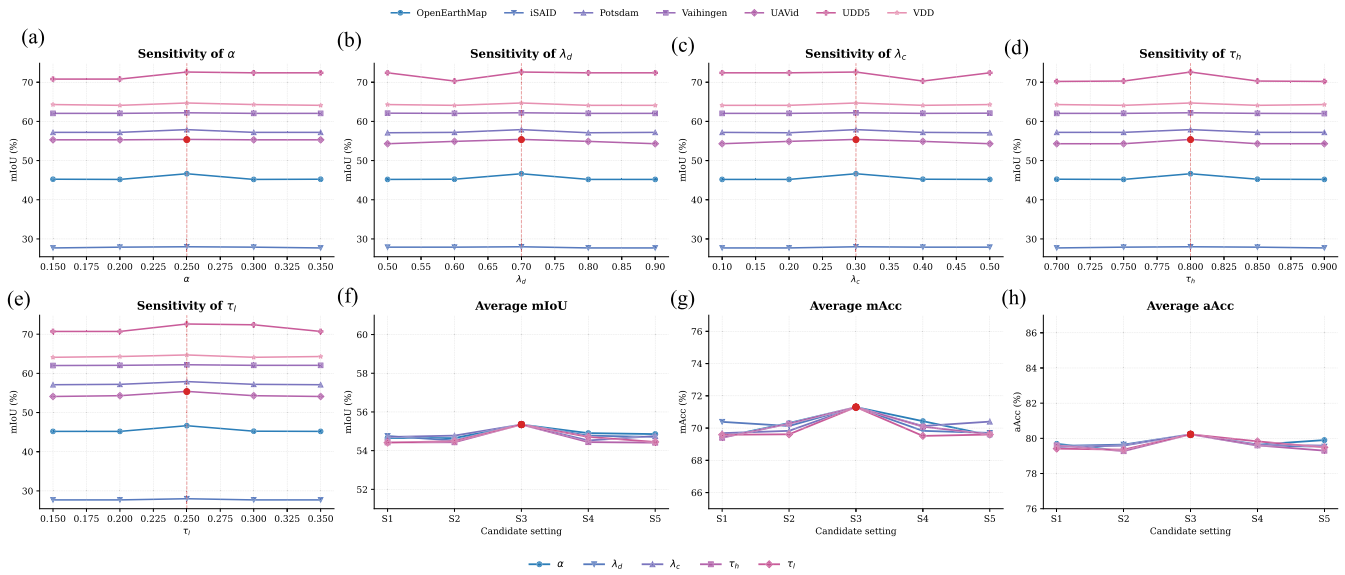


Fig. 11. Hyperparameter sensitivity analysis of GLoP-SAM3. (a)–(e) Sensitivity of key hyperparameters, including α, λ_d, λ_c, τ_h, and τ_l, on different datasets. (f)–(h) Average mIoU, mAcc, and aAcc under different candidate settings.

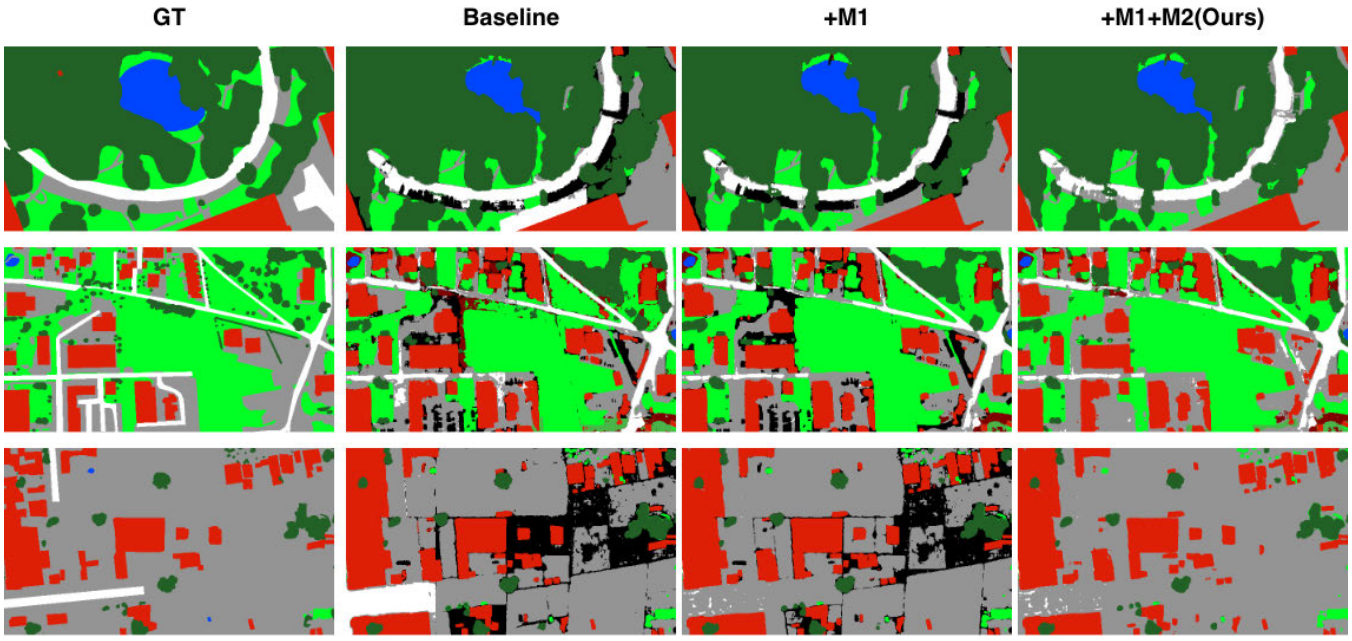


Fig. 12. Qualitative ablation results of the proposed modules. From (left to right) ground truth, baseline, +M1, and +M1 +M2 (Ours). Compared with the baseline, +M1 improves global semantic consistency and region completeness, while +M1 +M2 further suppresses unreliable prior propagation and produces more stable boundary and local-region predictions.

with the full-image response. For RGPR, τ_h is used to protect highly reliable local predictions from over-correction, while τ_l helps identify ambiguous regions that may benefit from global-prior enhancement. From the experimental perspective, the selected setting achieves the best or near-best performance across mIoU, mAcc, and aAcc, and the neighboring settings show only moderate performance variations. This indicates that the hyperparameter selection follows the principle of introducing global priors only when necessary and relatively reliable, while preserving locally confident structures.

2) Qualitative Analysis: To more intuitively demonstrate the effects of different modules on the prediction results, we further provide qualitative ablation results. Specifically, we compare the results of ground truth, baseline, +M1, and +M1 +M2, where baseline denotes using SAM3 alone as the basic open-vocabulary predictor, +M1 denotes adding the DGLF module, and +M1 +M2 denotes the complete GLoP-SAM3 framework.

As shown in Fig. 12, the baseline tends to suffer from fragmented local predictions, incomplete regions, and class confusion in complex remote sensing scenes. For example, in large vegetation, road, building, and water regions, the baseline results contain noticeable noise and discontinuities, and some areas are incorrectly assigned to adjacent

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categories. After introducing M1, the predictions are improved in terms of global semantic consistency and region completeness. This indicates that global–local prior fusion can exploit full-image contextual information to alleviate unstable local responses caused by sliding-window inference, making predictions within the same semantic region more coherent.

After further adding M2, the prediction results become more stable, and part of the error propagation caused by unreliable global priors is suppressed. In particular, around road boundaries, building edges, and small local regions, +M1 +M2 better preserves high-confidence local structures and reduces error diffusion in ambiguous regions. Compared with +M1, the complete model produces more reliable results in terms of boundary integrity, local-region stability, and noise suppression, making the predictions closer to the ground truth.

Overall, the qualitative ablation results confirm the complementary roles of the two modules. M1 improves region-level semantic consistency by introducing global priors, while M2 suppresses unreliable prior propagation through reliability gating. This is consistent with the quantitative ablation results, showing that DGLF and RGPR jointly enhance the segmentation stability of GLoP-SAM3 in complex remote sensing scenes.

V. Discussion and Conclusion

In this article, we propose GLoP-SAM3, a dual post-processing framework for open-vocabulary remote sensing semantic segmentation, to address the limitations of unstable local predictions, insufficient global semantic consistency, and unreliable prior propagation in SegEarth-OV3. Specifically, the proposed divergence-aware global–local prior fusion module improves semantic consistency by adaptively integrating full-image priors with local predictions, while the reliability-gated refinement module further enhances ambiguous regions and suppresses unreliable prior propagation. Extensive experiments on multiple representative datasets show that GLoP-SAM3 achieves consistent improvements over strong baselines and existing training-free methods, with only limited additional inference overhead. Although the method still shows certain limitations on challenging datasets such as LoveDA, the overall results demonstrate the effectiveness, robustness, and practical value of the proposed framework for open-vocabulary remote sensing semantic segmentation.

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